

BFS Traversal

Step 1: Start at A

Legend

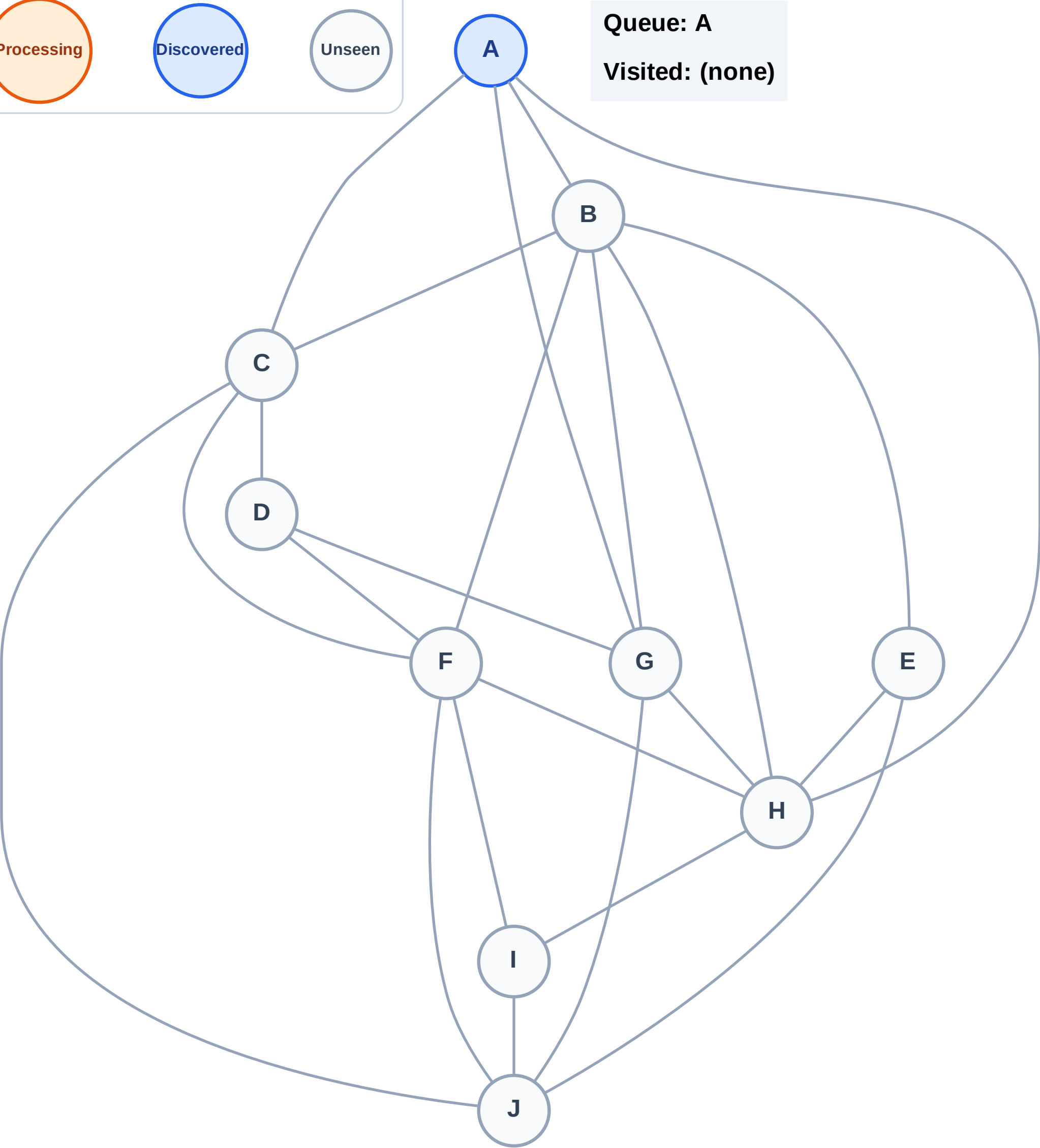
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

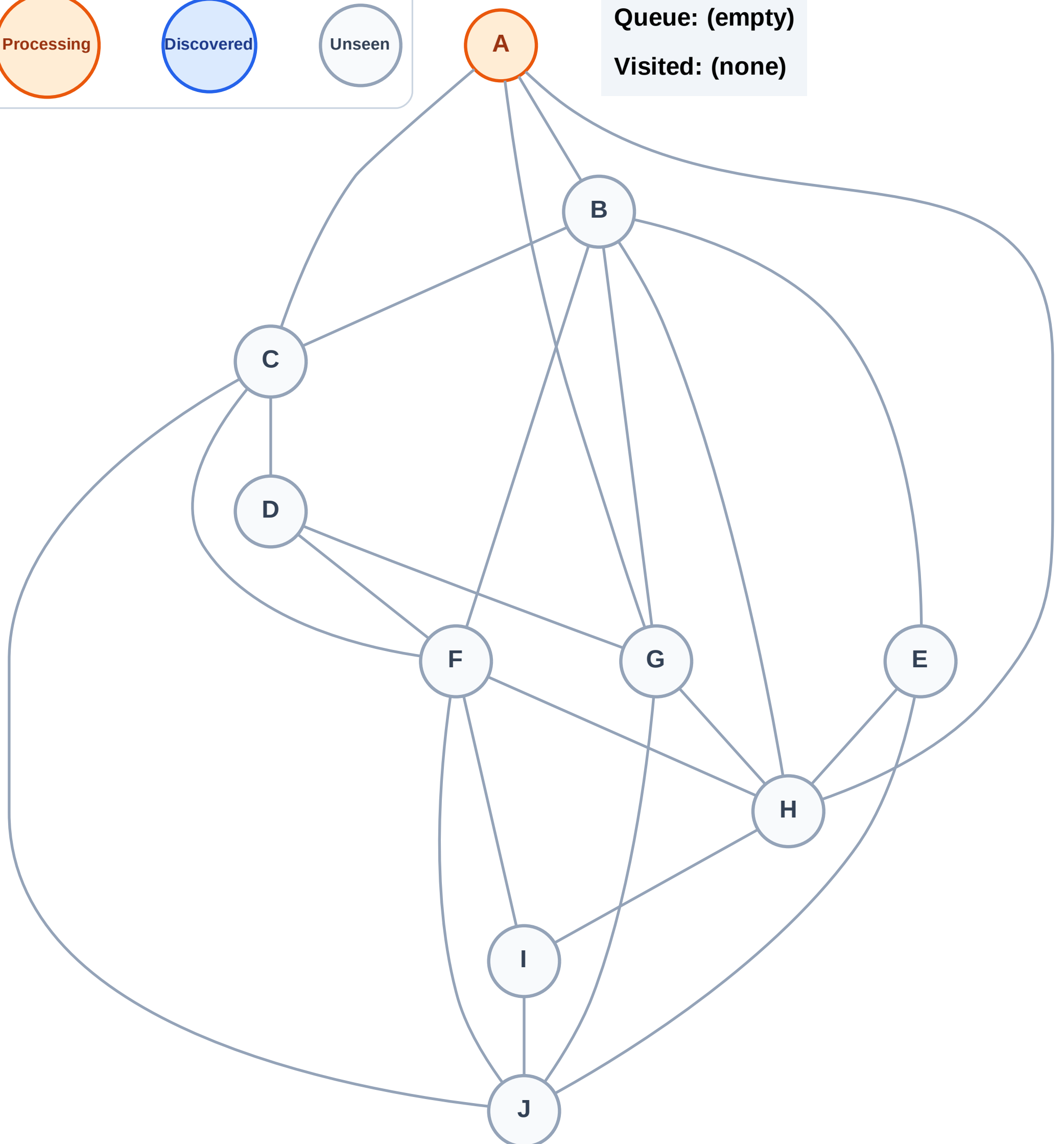
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

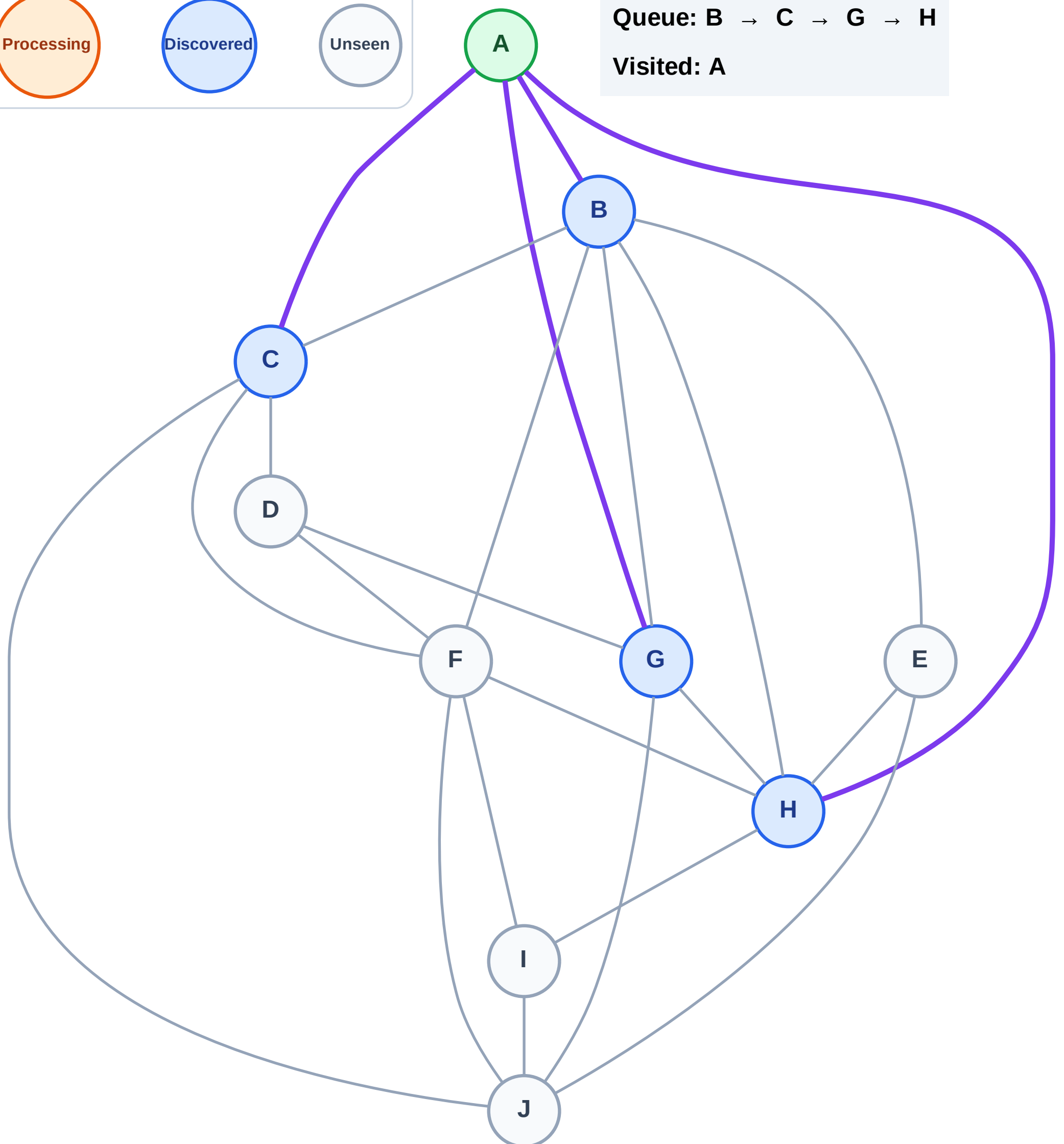
Step 3: Discover B, C, G, H from A

Legend



Queue: B → C → G → H

Visited: A



BFS Traversal

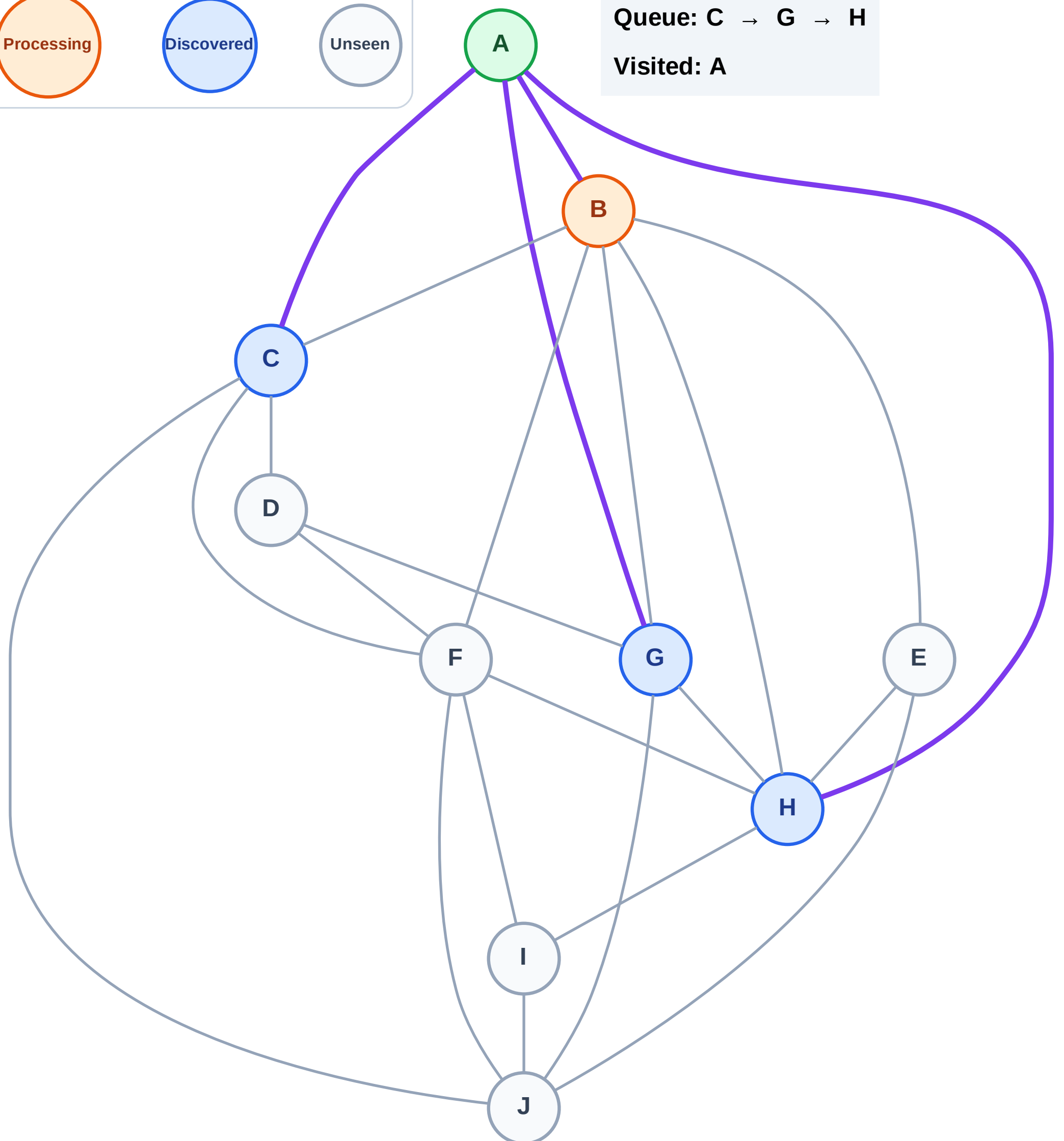
Step 4: Process node B

Legend



Queue: C → G → H

Visited: A



BFS Traversal

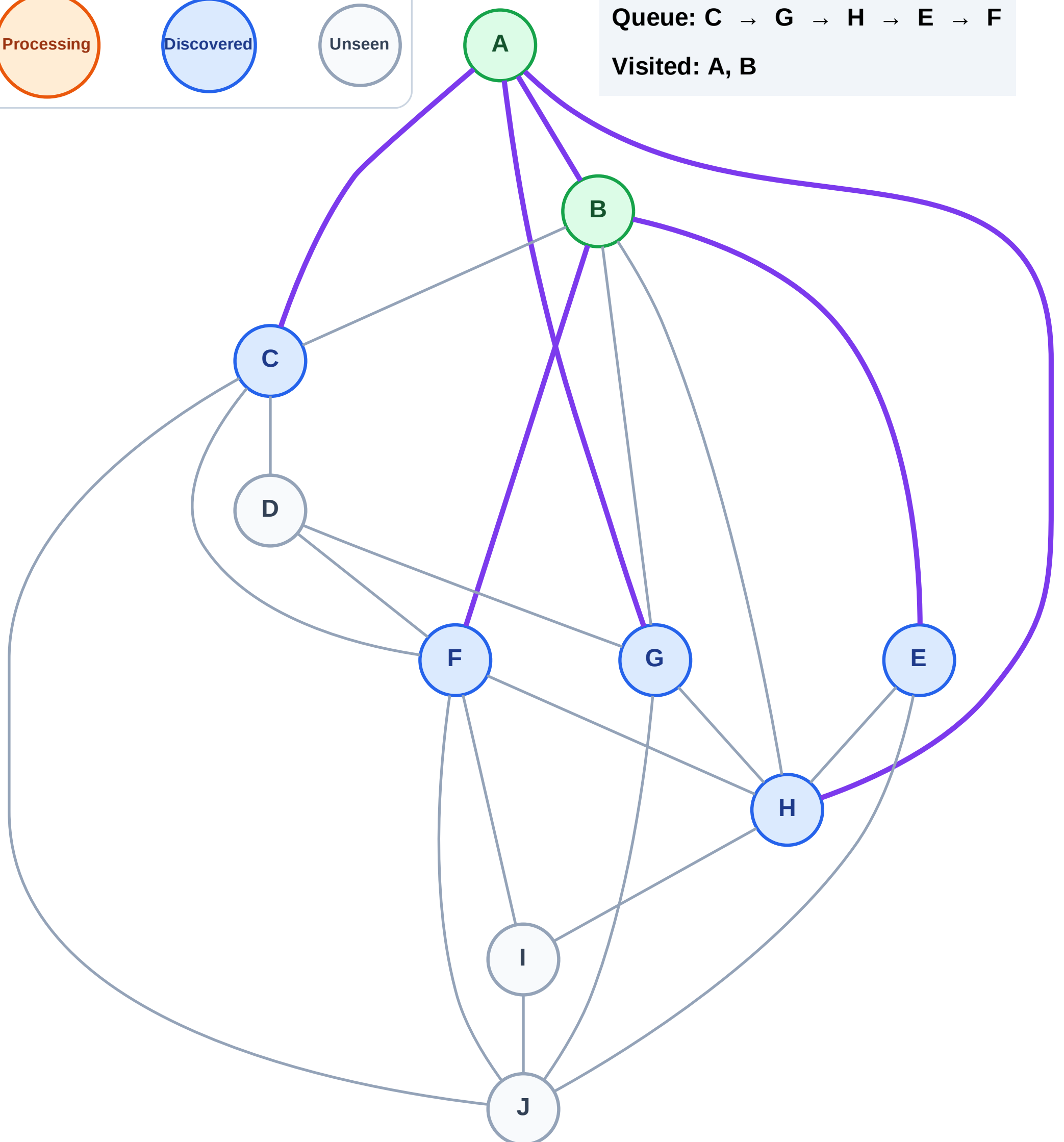
Step 5: Discover E, F from B

Legend



Queue: C → G → H → E → F

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

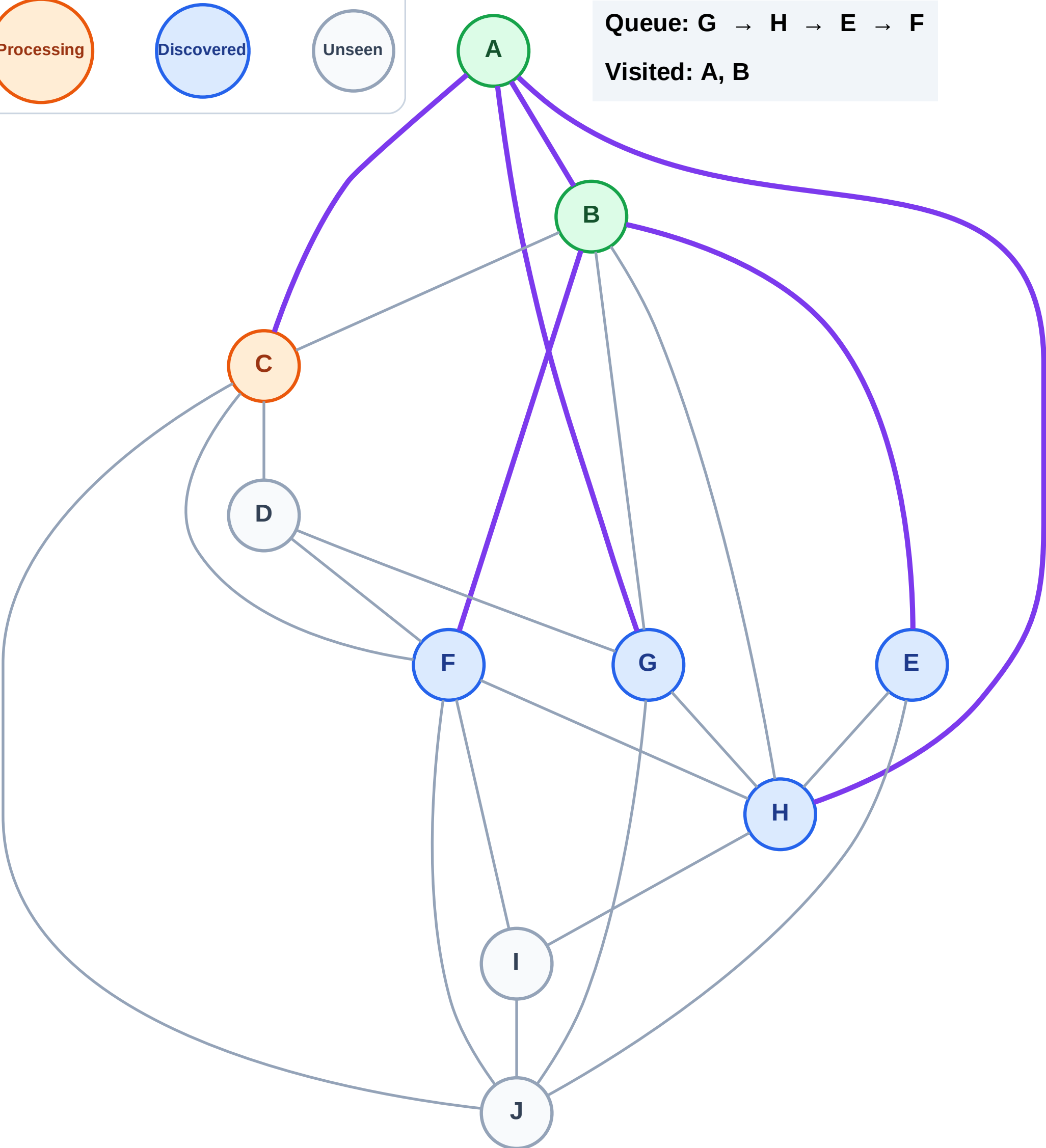
Processing

Discovered

Unseen

Queue: G → H → E → F

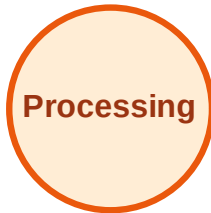
Visited: A, B



BFS Traversal

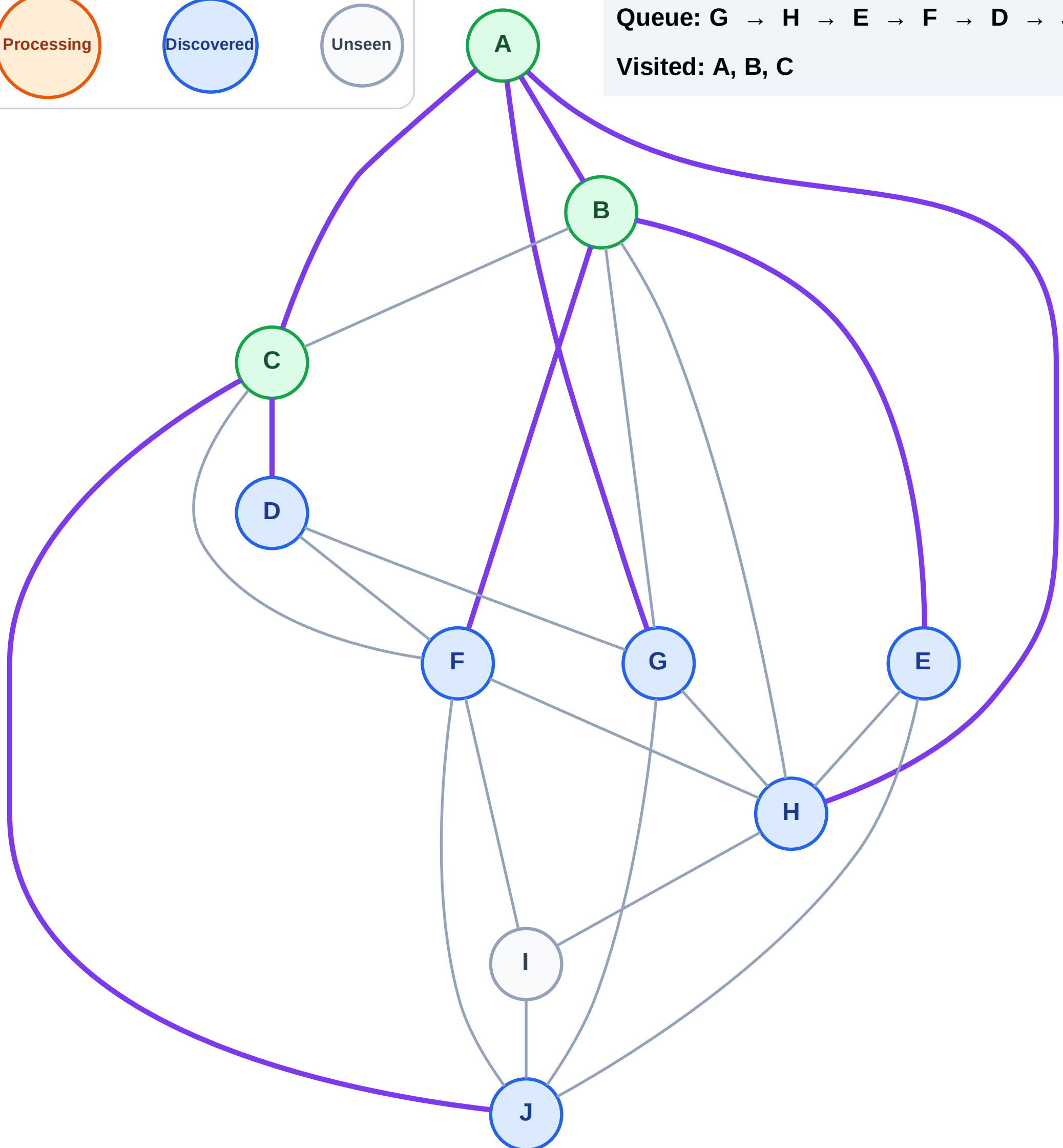
Step 7: Discover D, J from C

Legend



Queue: G → H → E → F → D → J

Visited: A, B, C



BFS Traversal

Step 8: Process node G

Legend

Complete

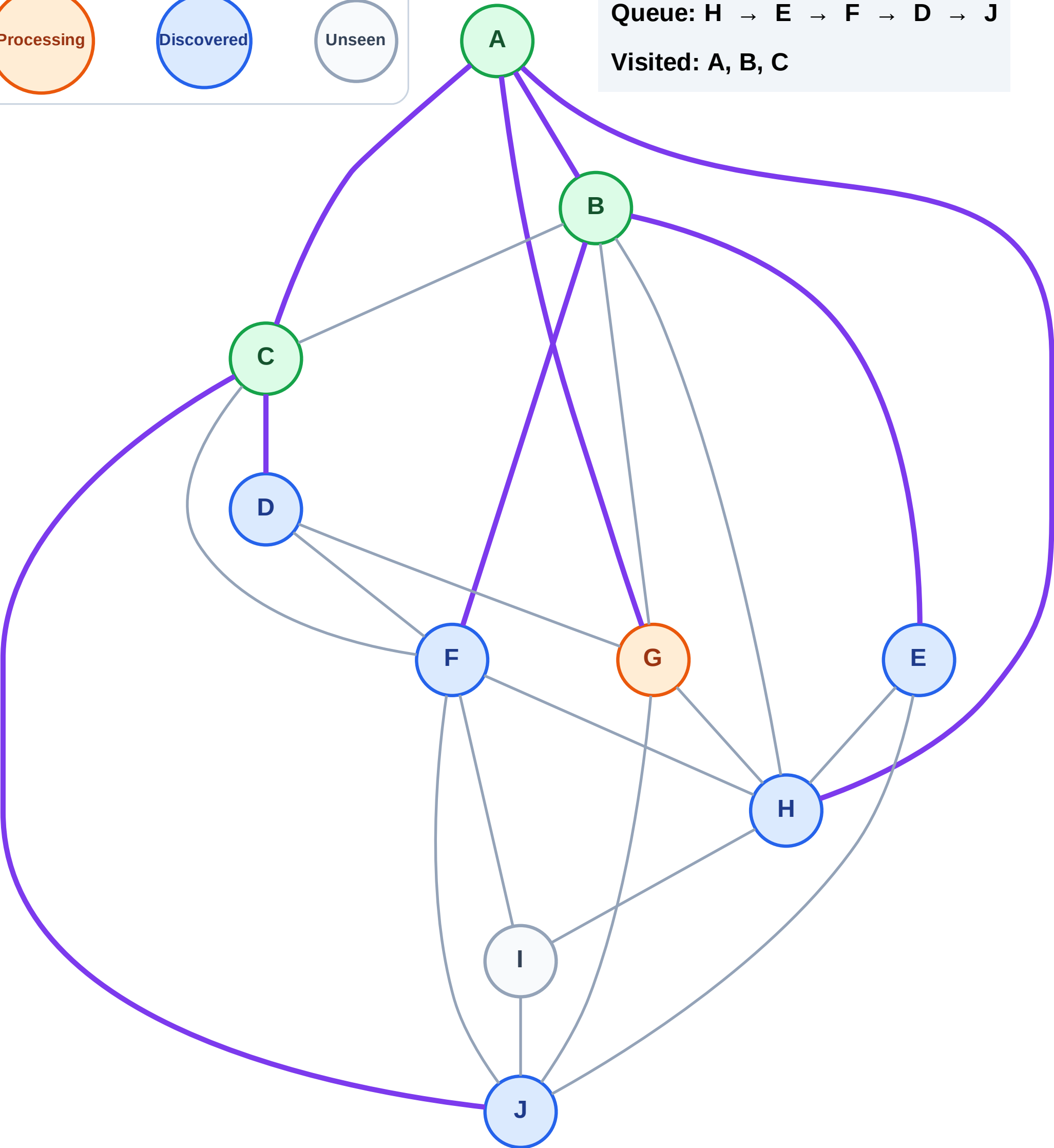
Processing

Discovered

Unseen

Queue: H → E → F → D → J

Visited: A, B, C



BFS Traversal

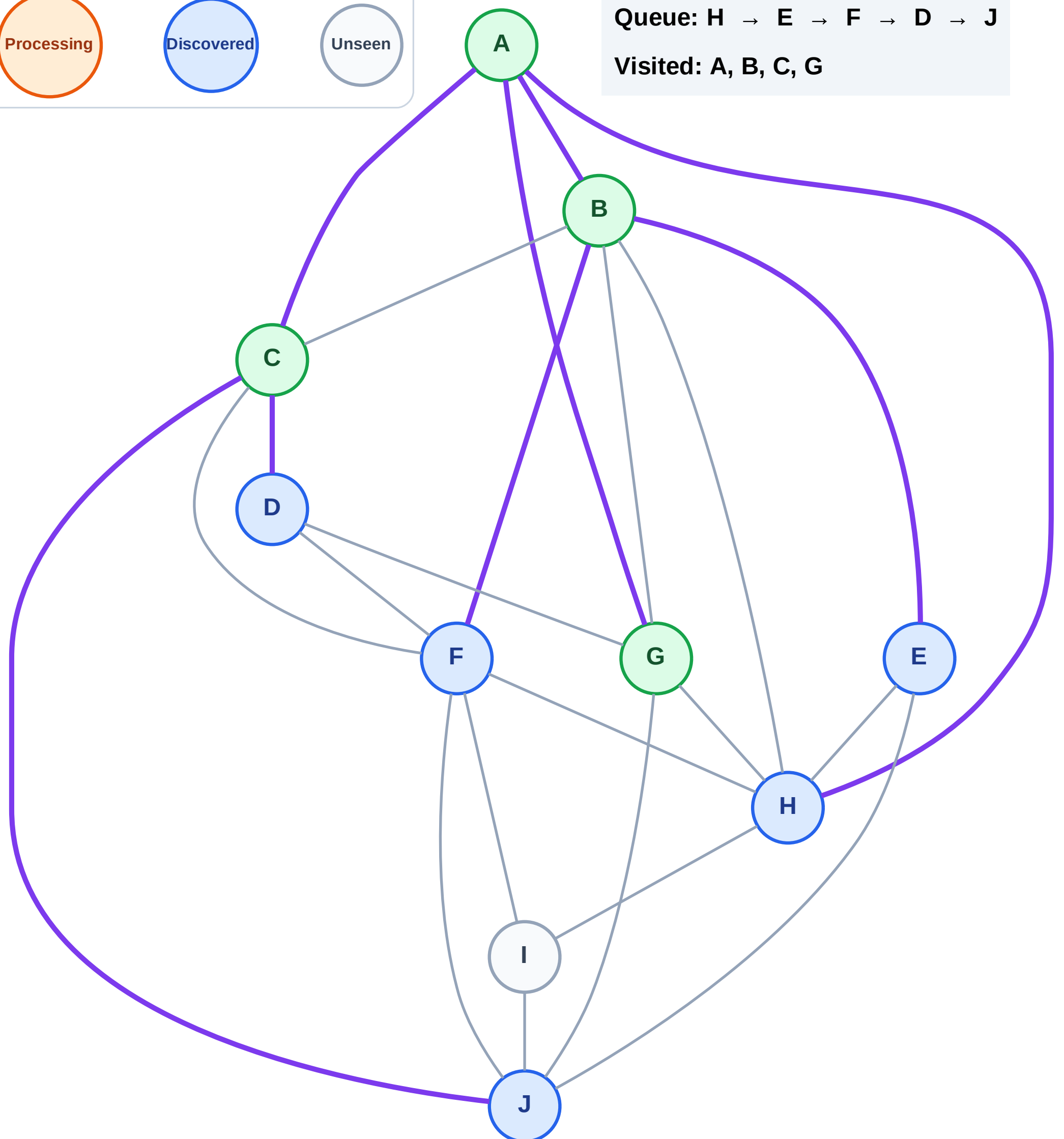
Step 9: No new neighbors from G

Legend



Queue: H → E → F → D → J

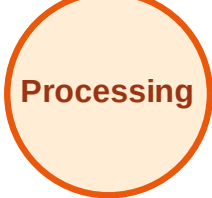
Visited: A, B, C, G



BFS Traversal

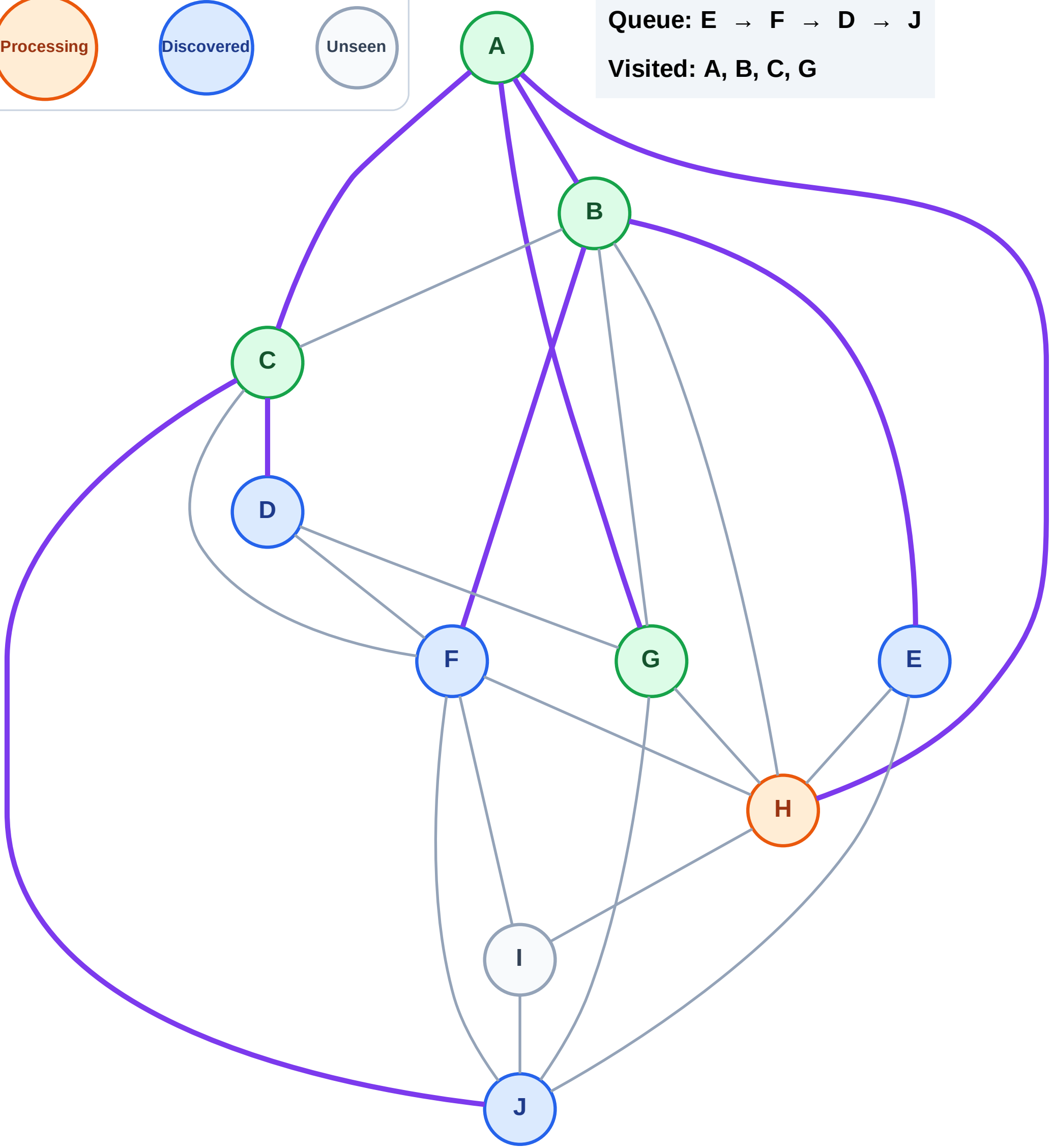
Step 10: Process node H

Legend



Queue: E → F → D → J

Visited: A, B, C, G



BFS Traversal

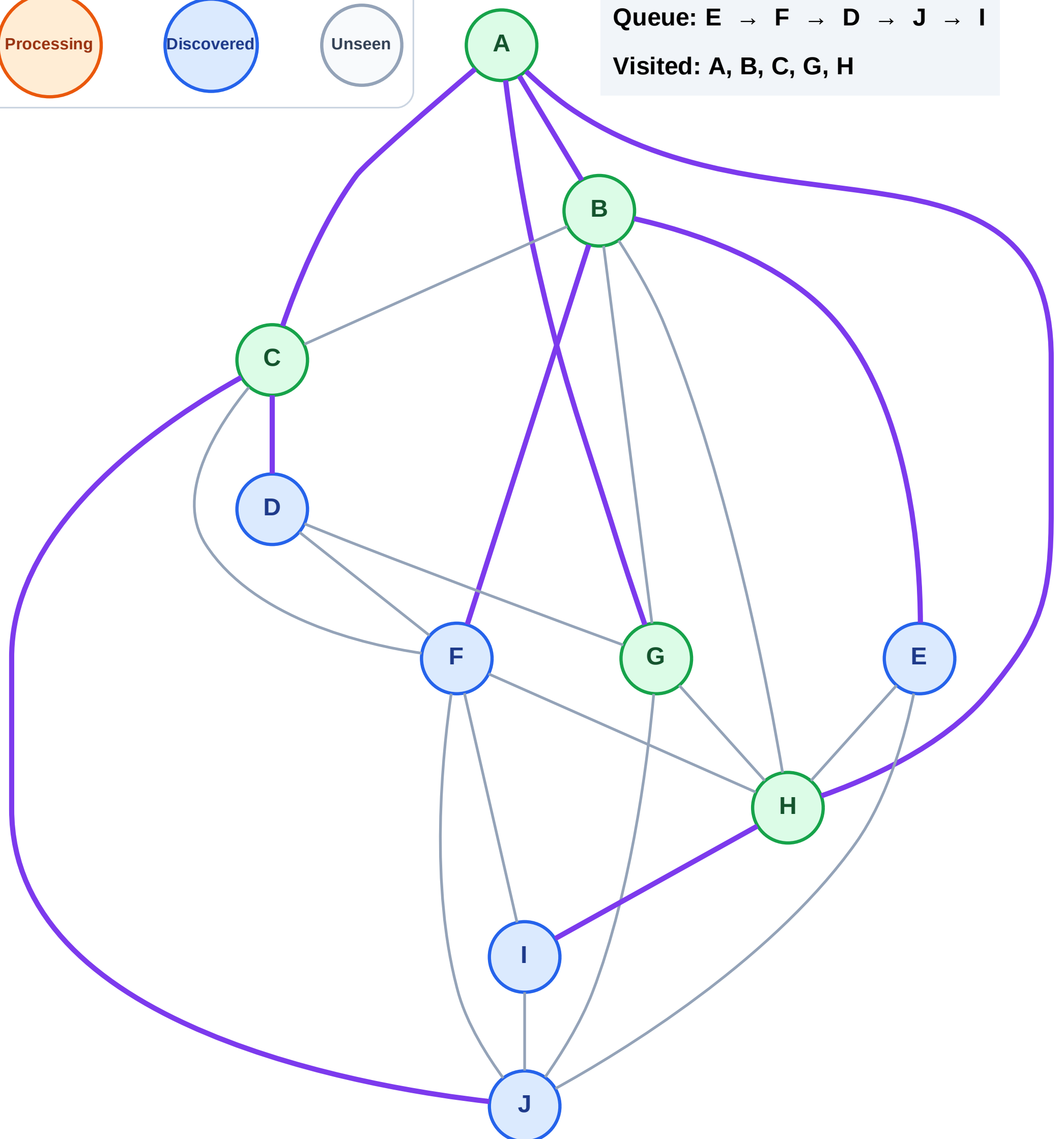
Step 11: Discover I from H

Legend



Queue: E → F → D → J → I

Visited: A, B, C, G, H



BFS Traversal

Step 12: Process node E

Legend

Complete

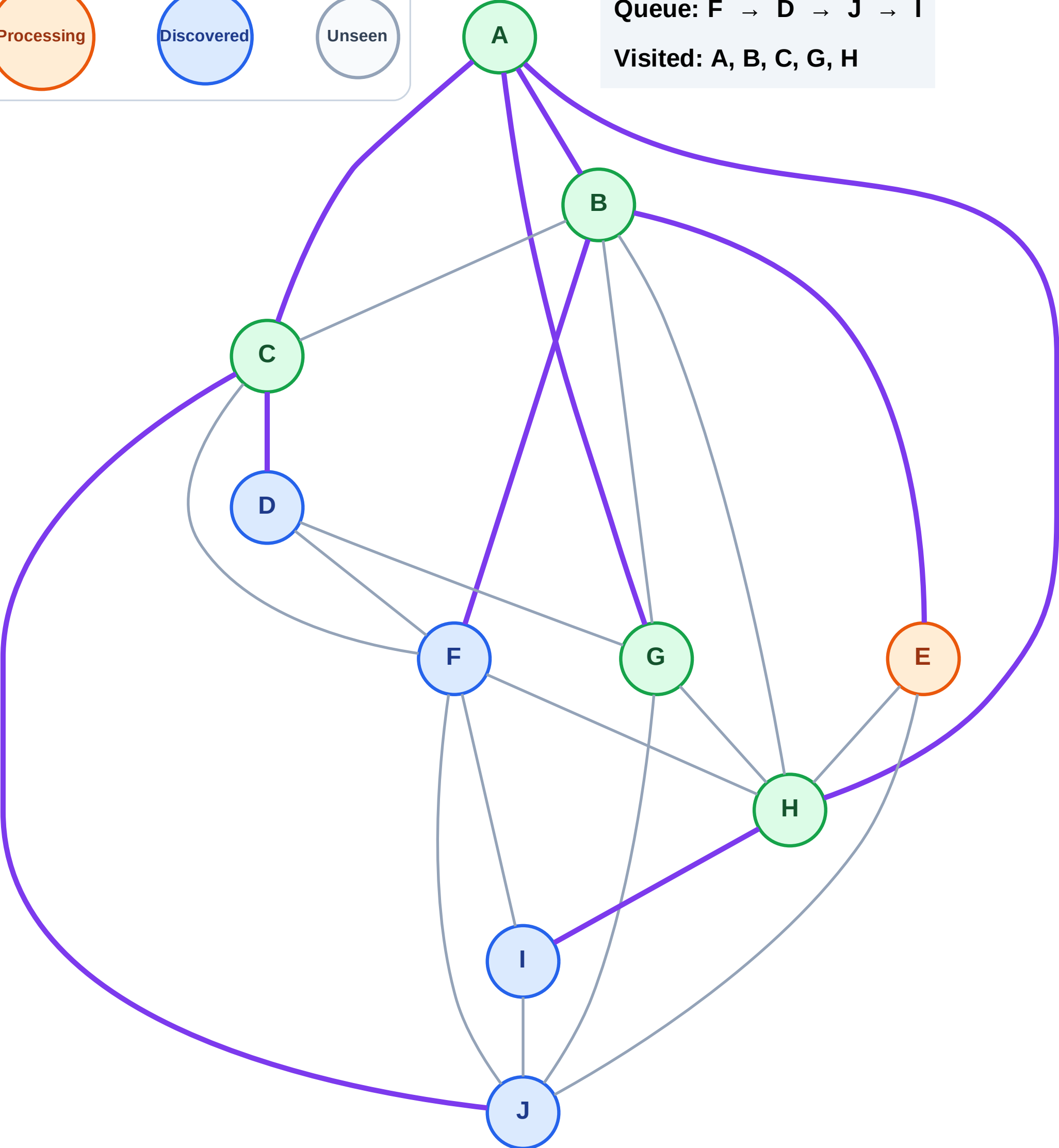
Processing

Discovered

Unseen

Queue: F → D → J → I

Visited: A, B, C, G, H



BFS Traversal

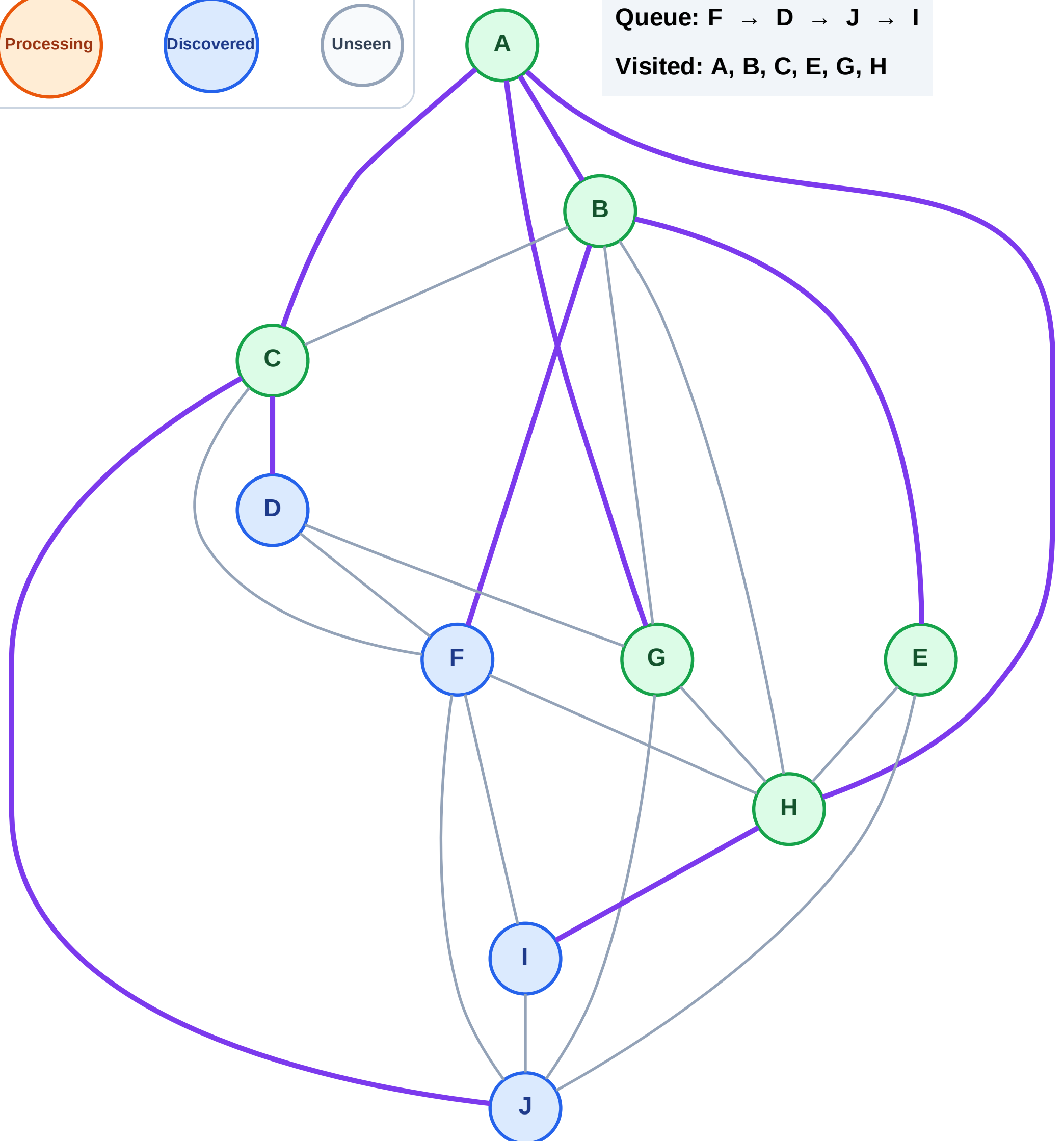
Step 13: No new neighbors from E

Legend



Queue: F → D → J → I

Visited: A, B, C, E, G, H



BFS Traversal

Step 14: Process node F

Legend

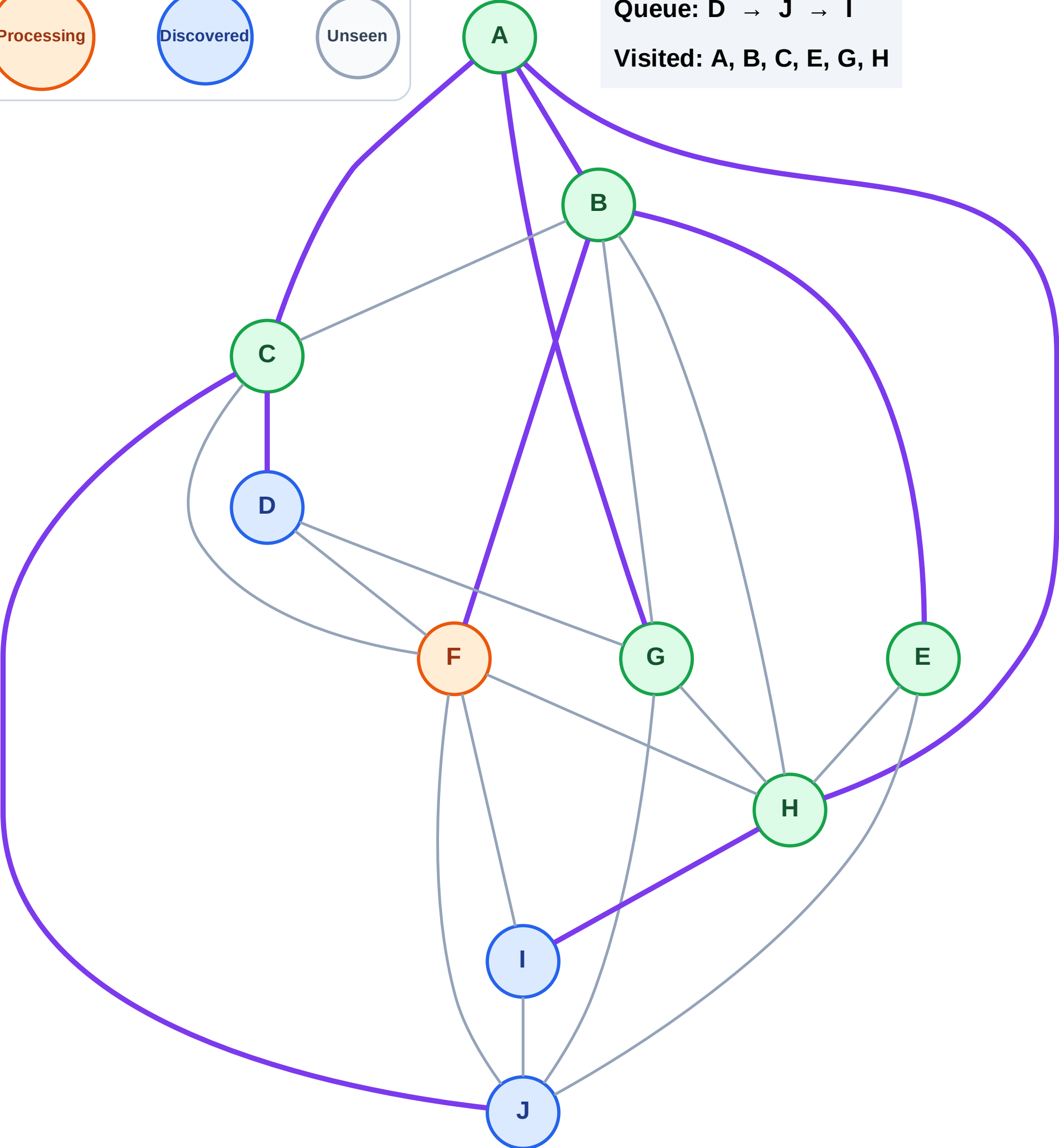
Complete

Processing

Discovered

Unseen

Queue: D → J → I
Visited: A, B, C, E, G, H



BFS Traversal

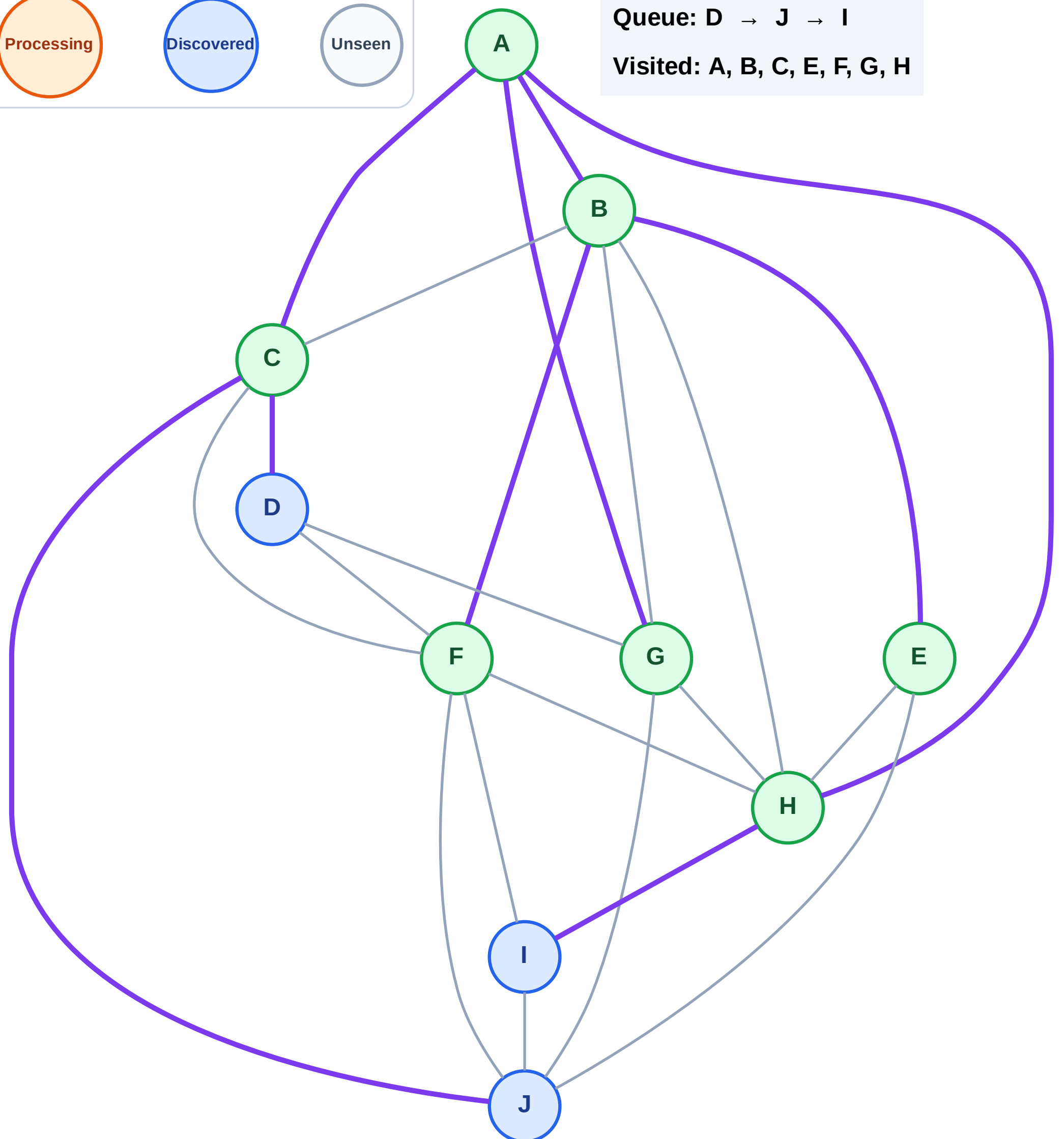
Step 15: No new neighbors from F

Legend



Queue: D → J → I

Visited: A, B, C, E, F, G, H



BFS Traversal

Step 16: Process node D

Legend

Complete

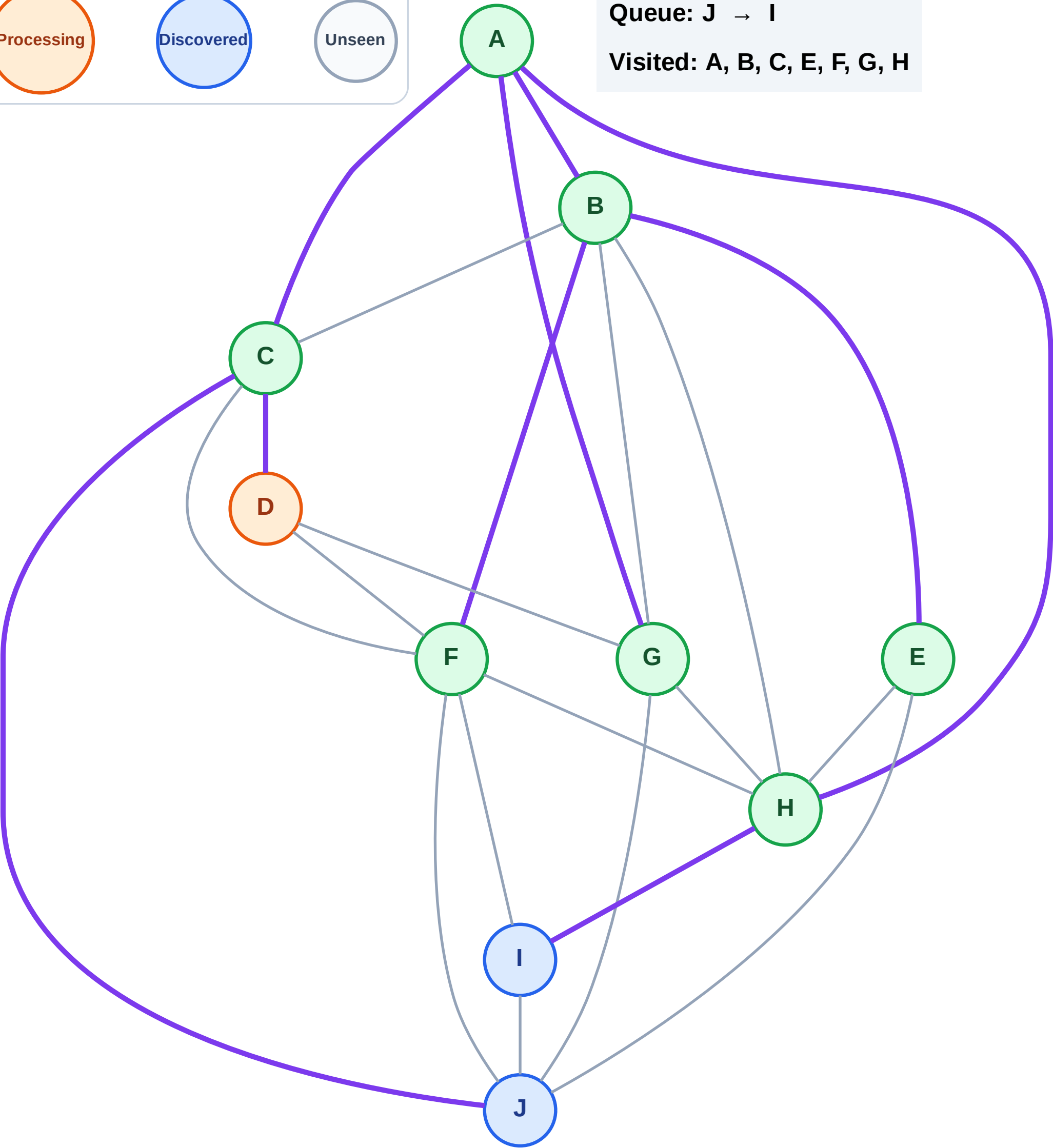
Processing

Discovered

Unseen

Queue: J → I

Visited: A, B, C, E, F, G, H



BFS Traversal

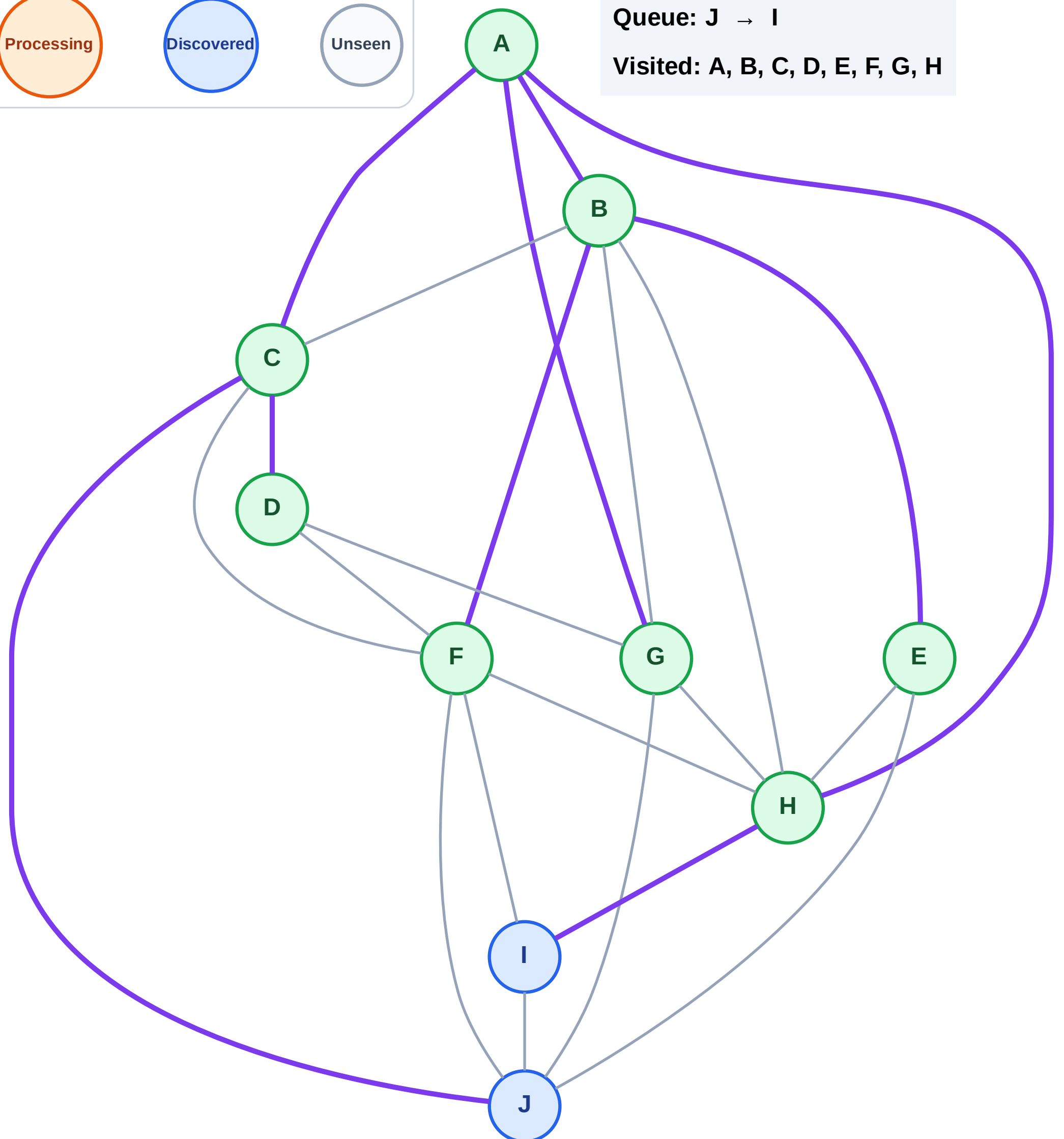
Step 17: No new neighbors from D

Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node J

Legend

Complete

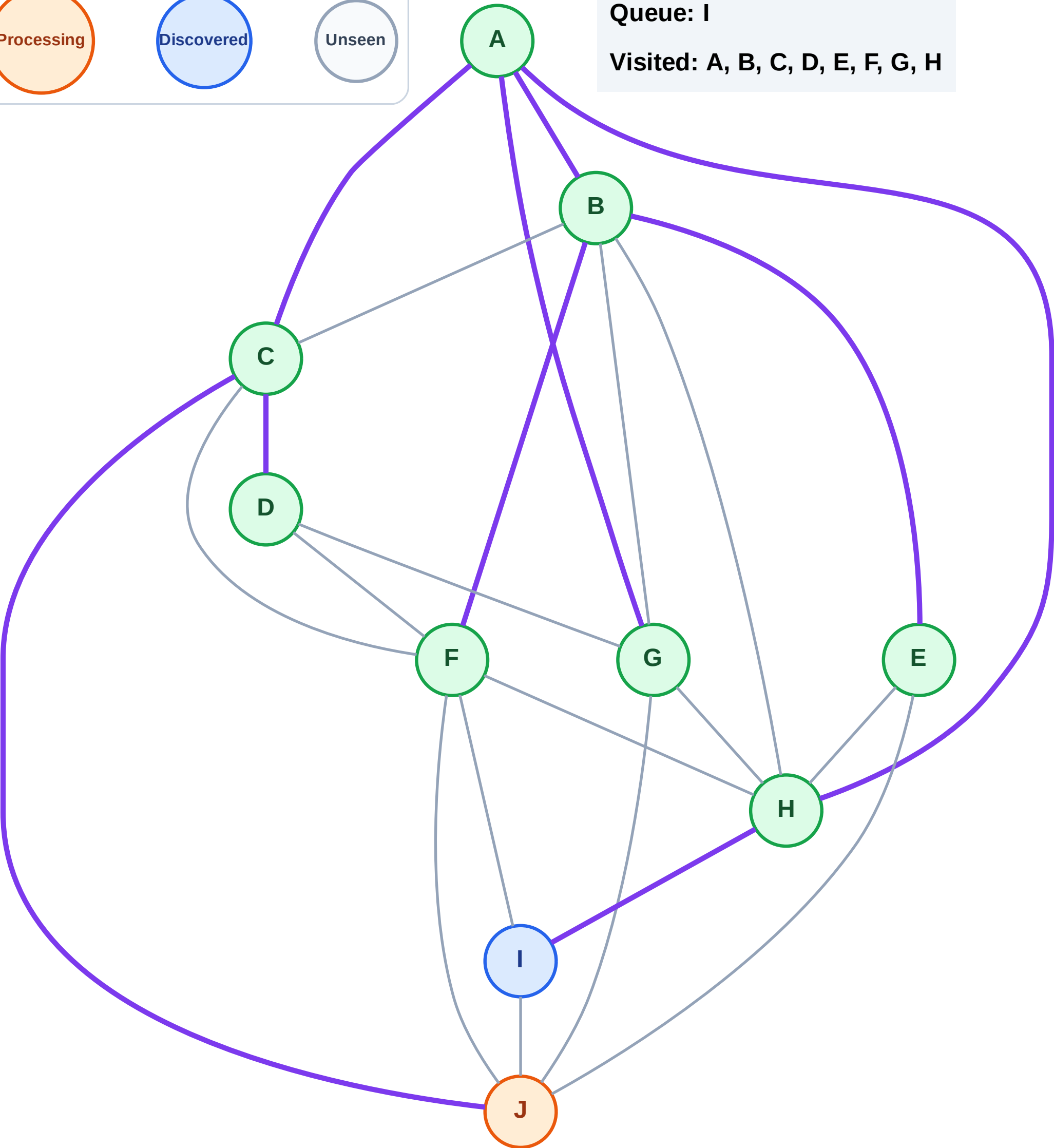
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, F, G, H



BFS Traversal

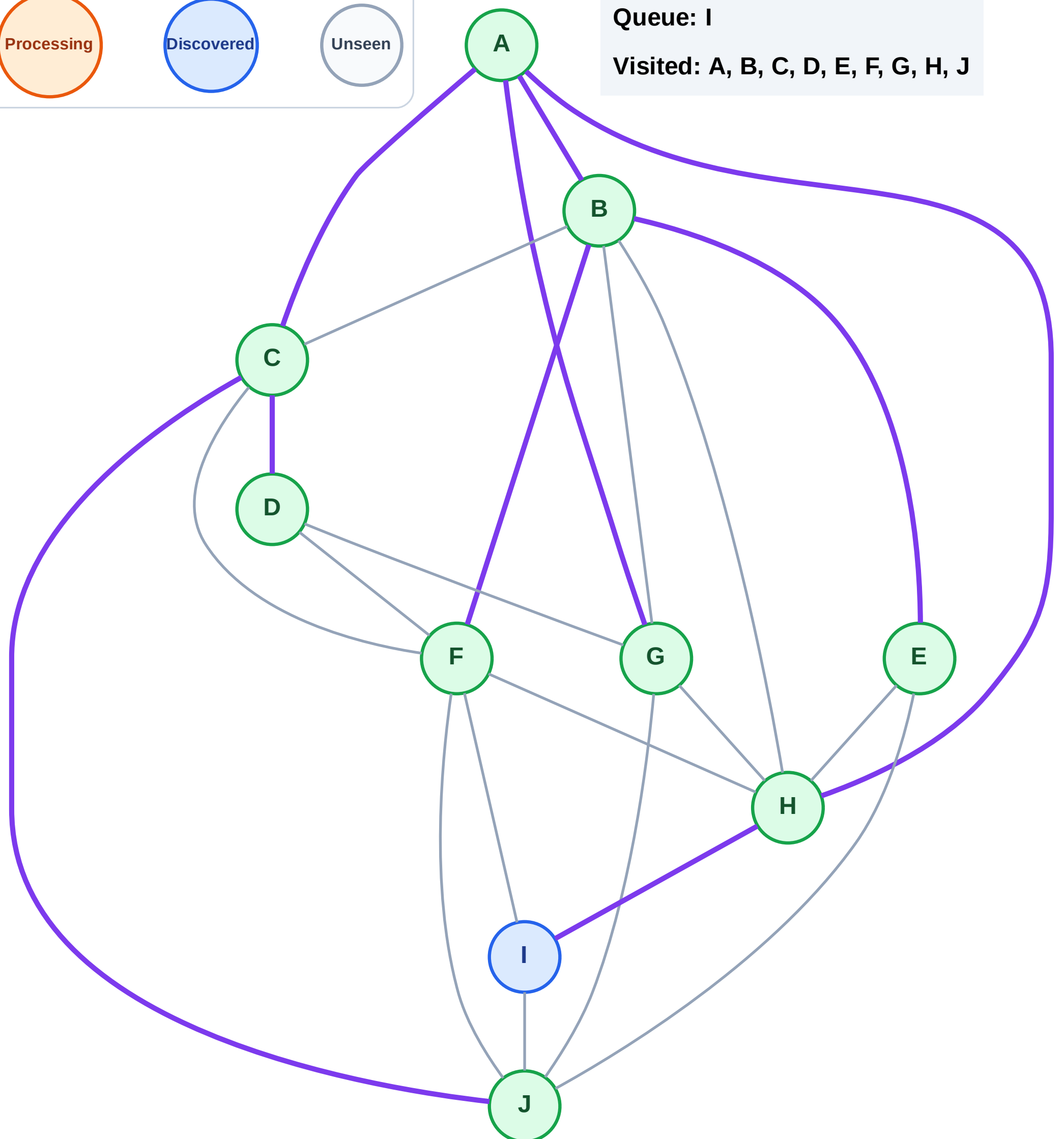
Step 19: No new neighbors from J

Legend



Queue: I

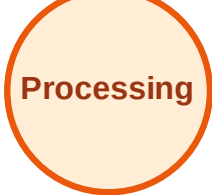
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

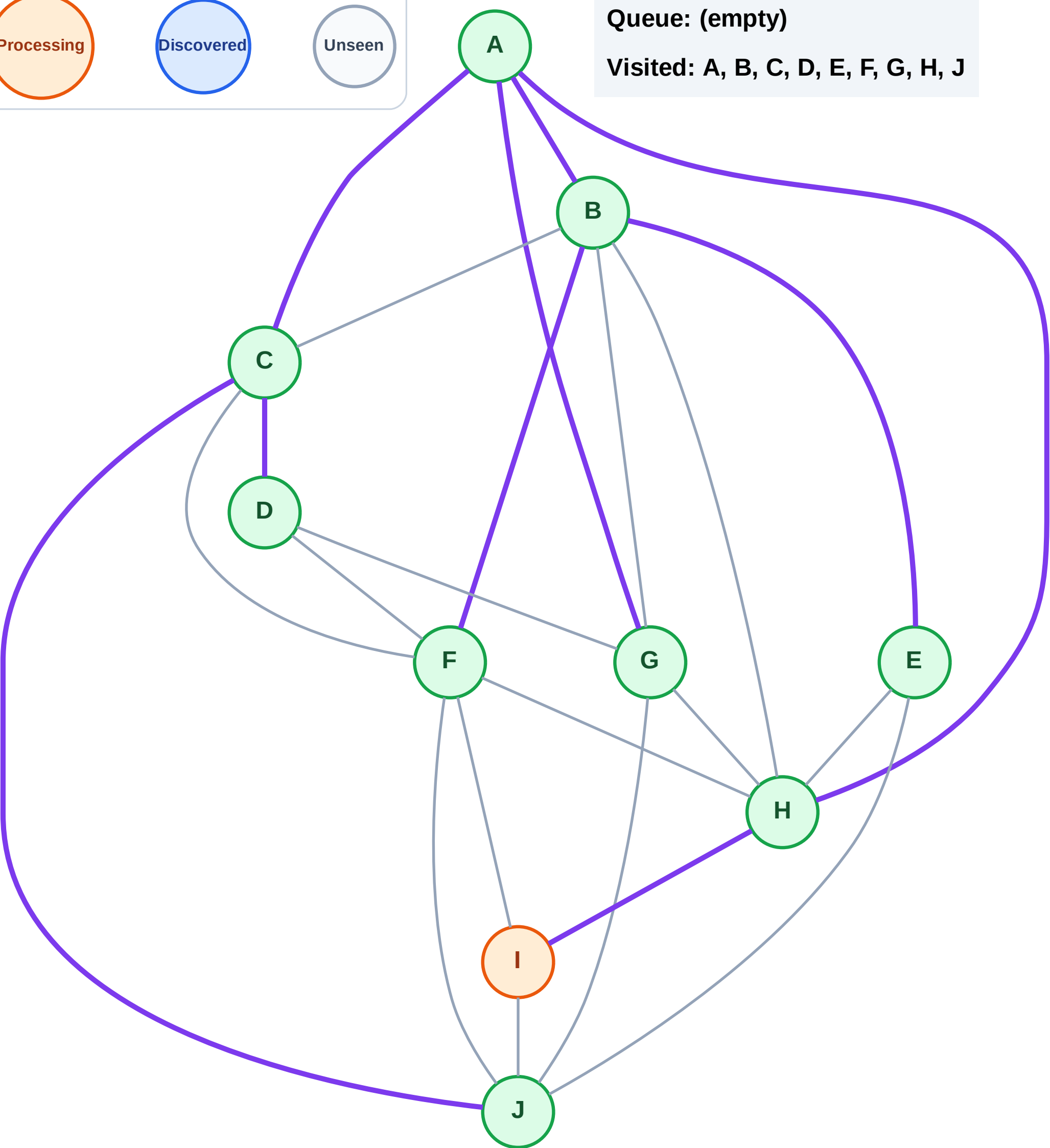
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

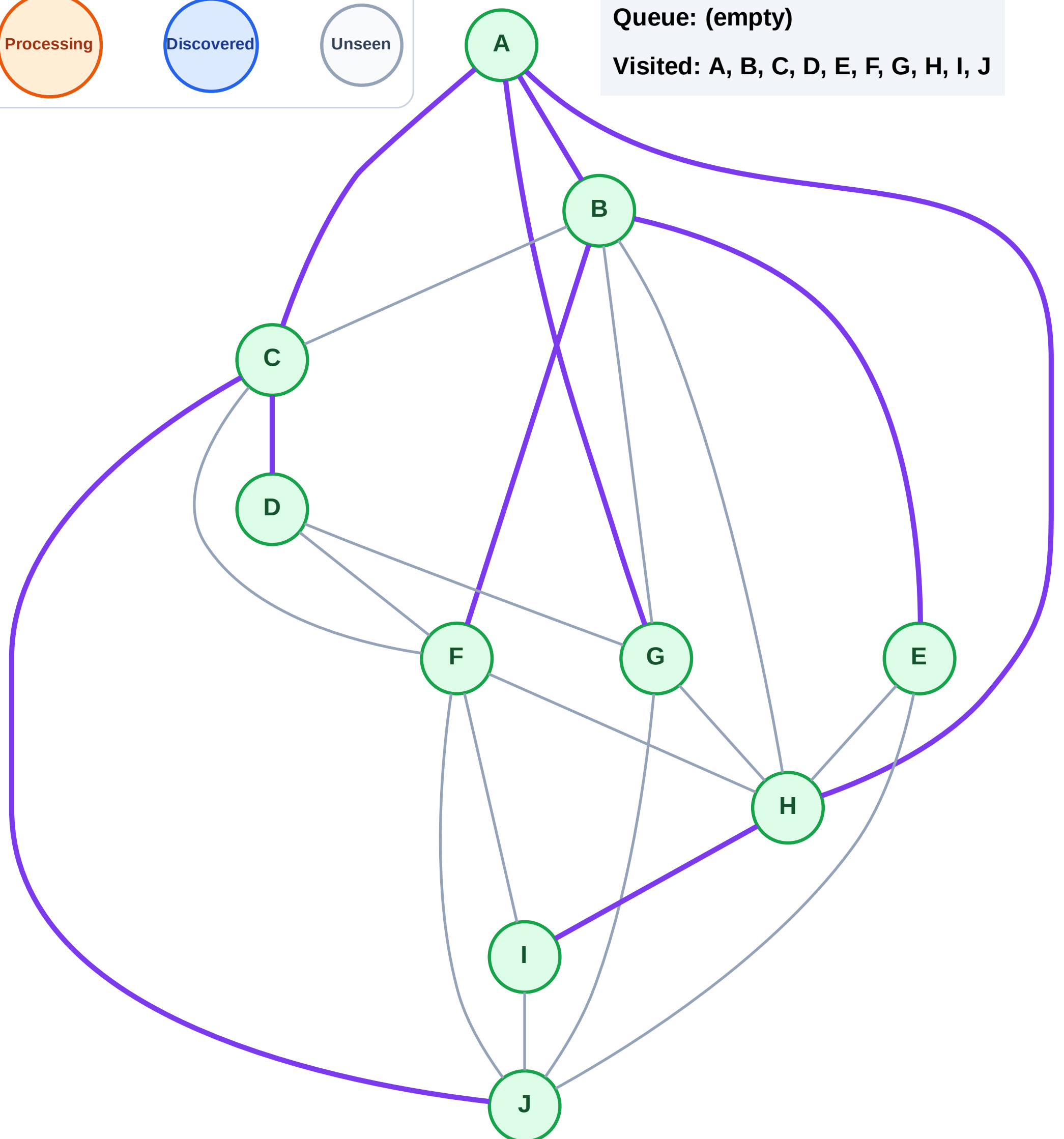
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

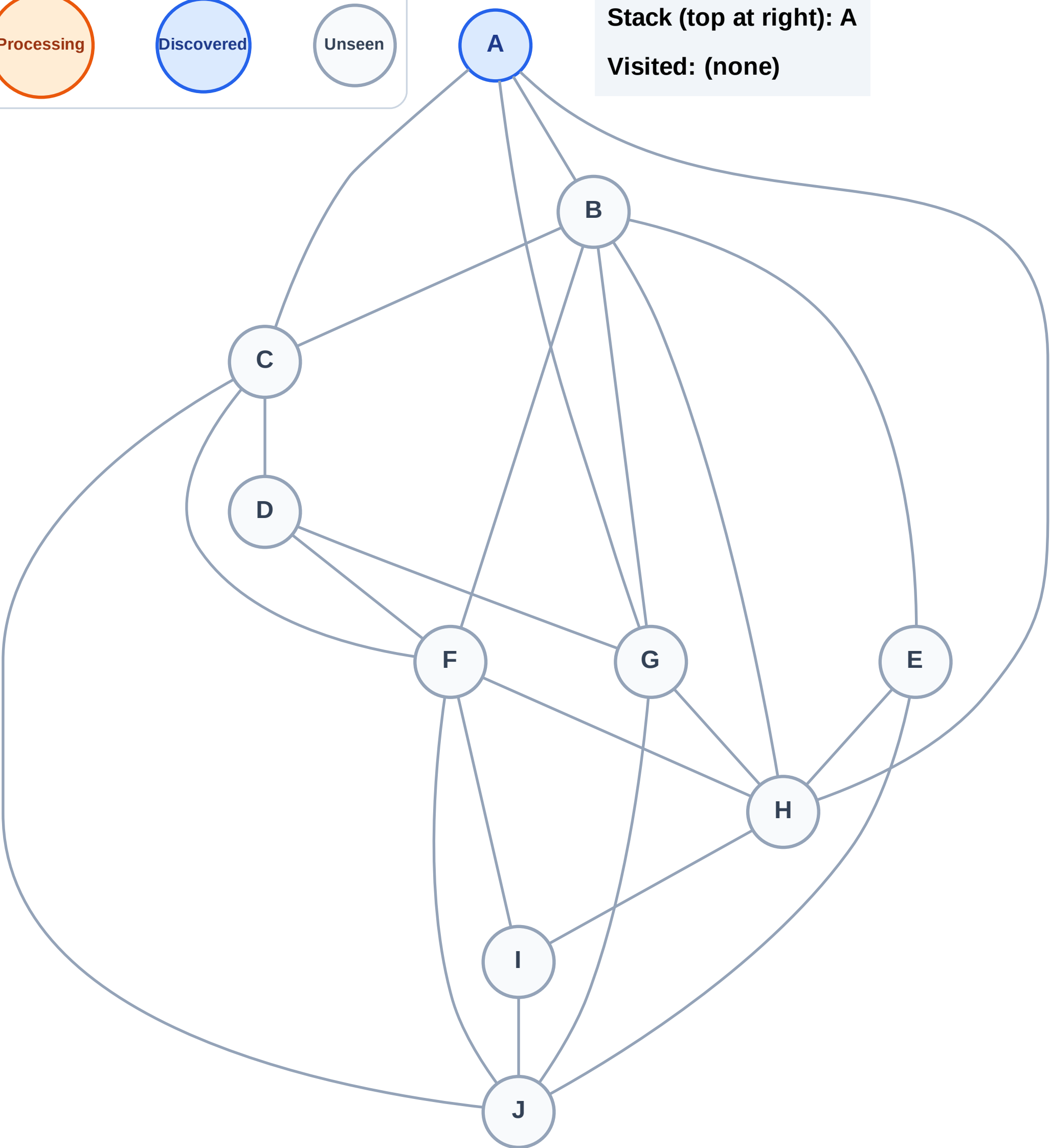
Processing

Discovered

Unseen

Stack (top at right): A

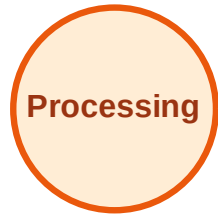
Visited: (none)



DFS Traversal

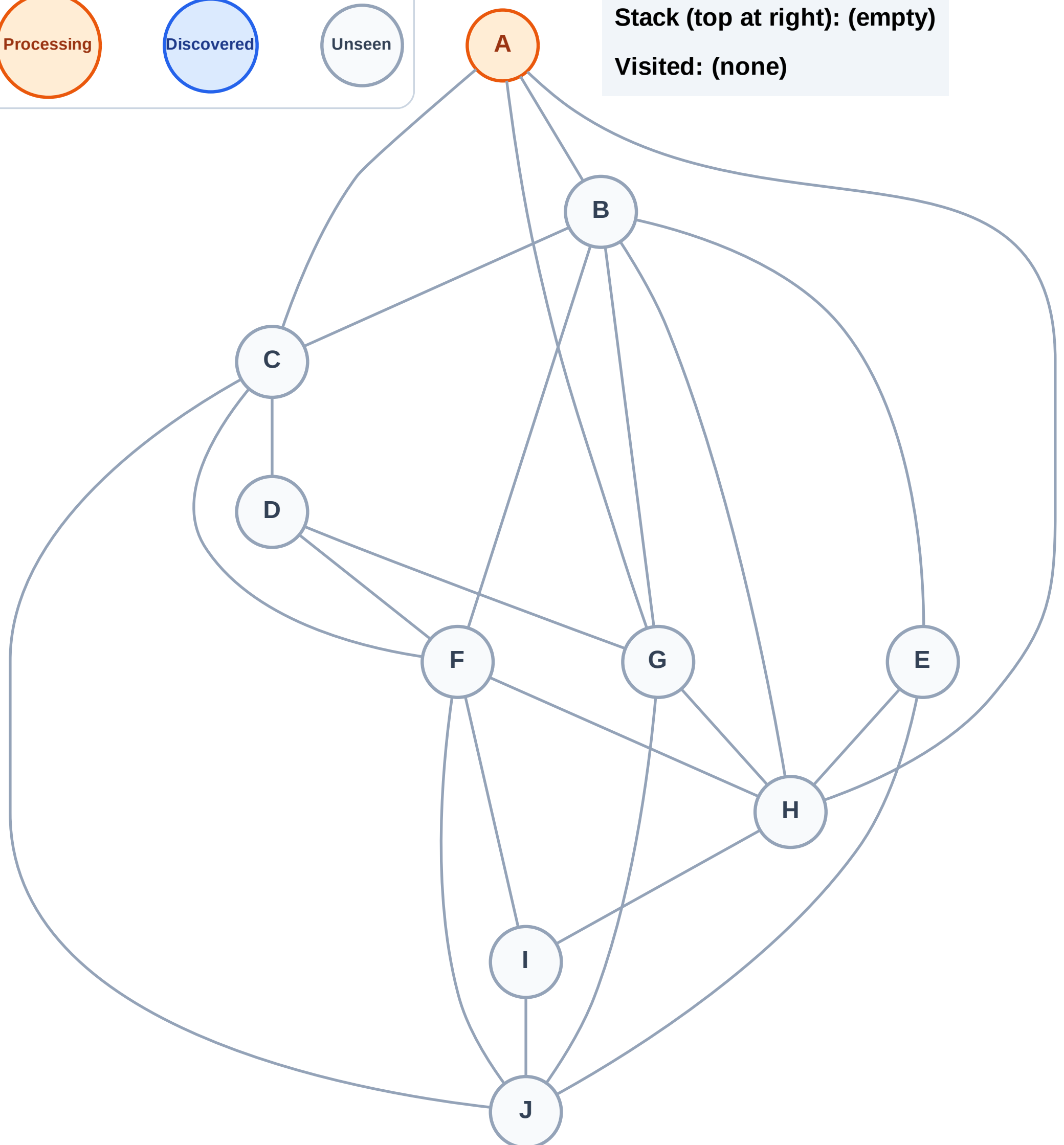
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

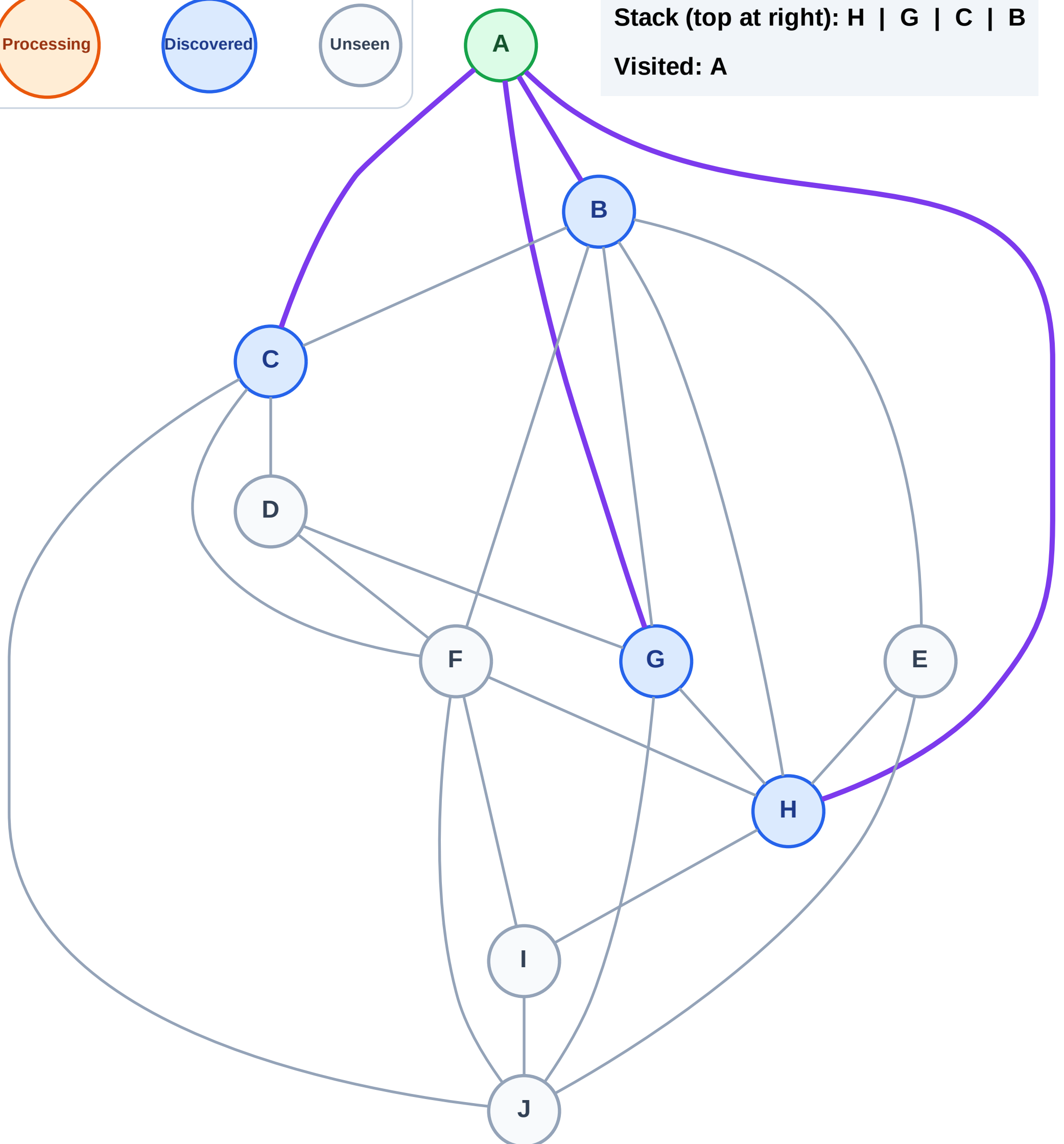
Step 3: Discover H, G, C, B from A

Legend



Stack (top at right): H | G | C | B

Visited: A



DFS Traversal

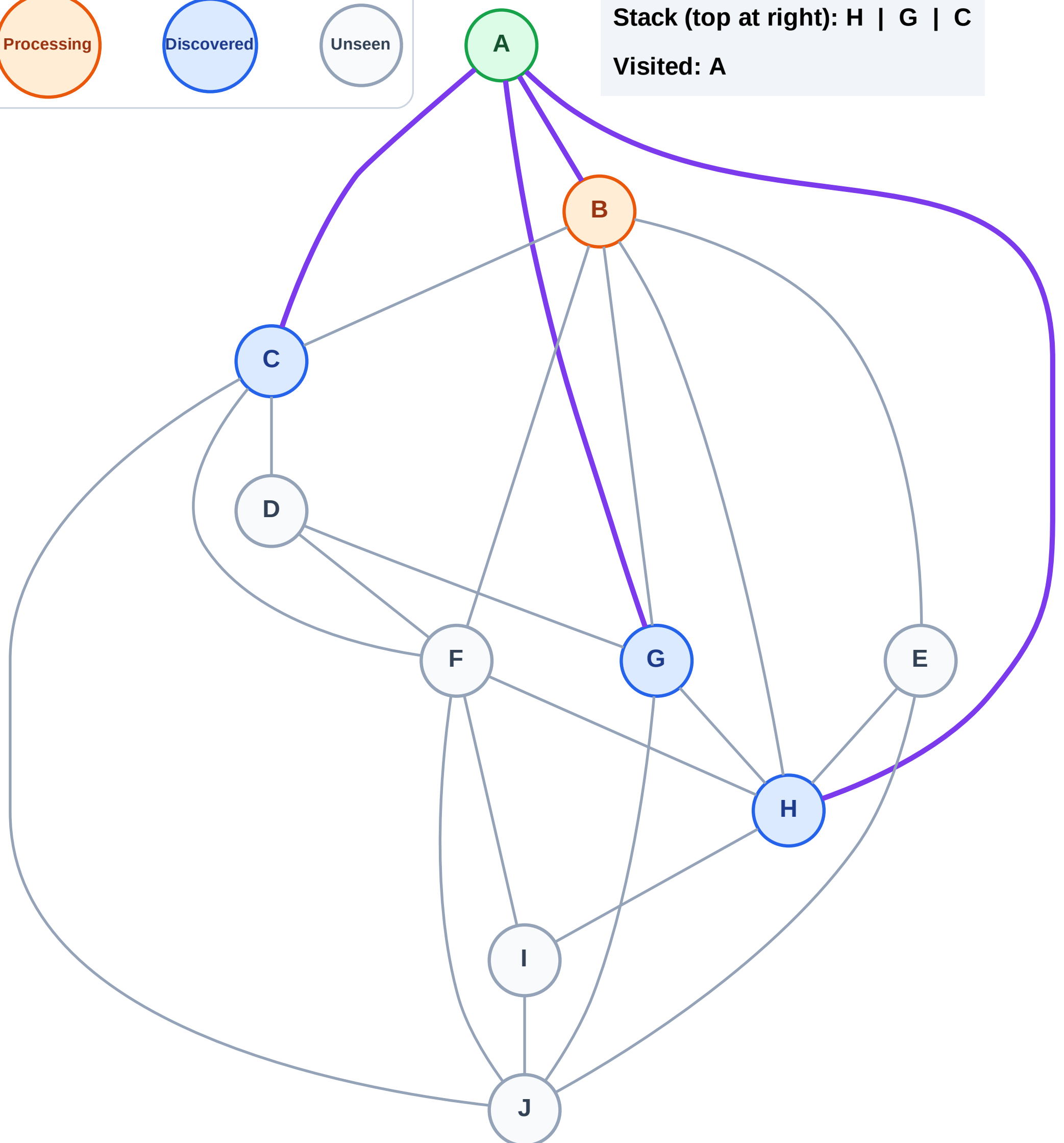
Step 4: Process node B

Legend



Stack (top at right): H | G | C

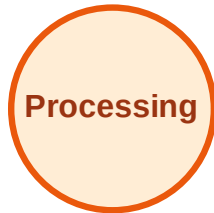
Visited: A



DFS Traversal

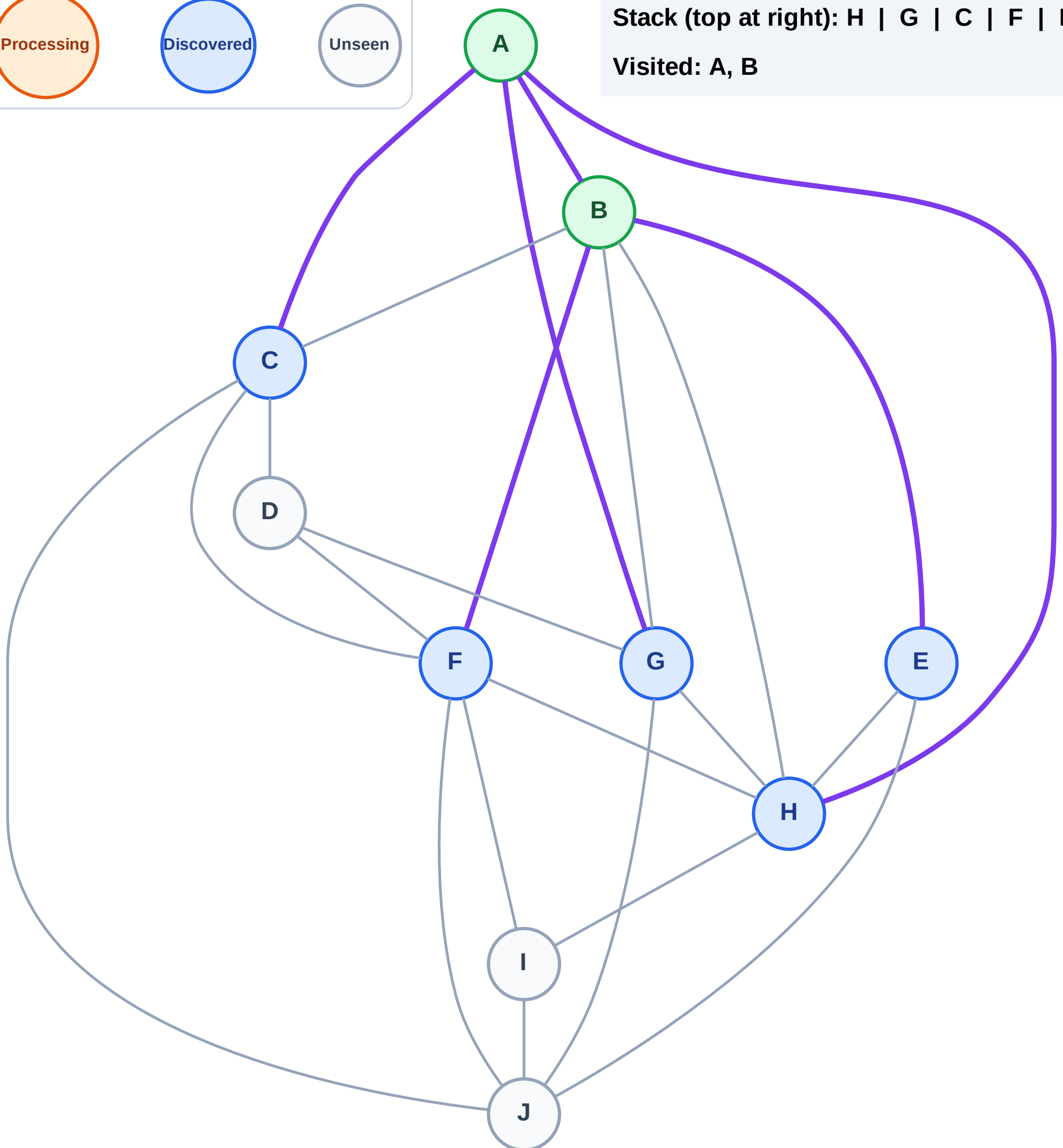
Step 5: Discover F, E from B

Legend



Stack (top at right): H | G | C | F | E

Visited: A, B



DFS Traversal

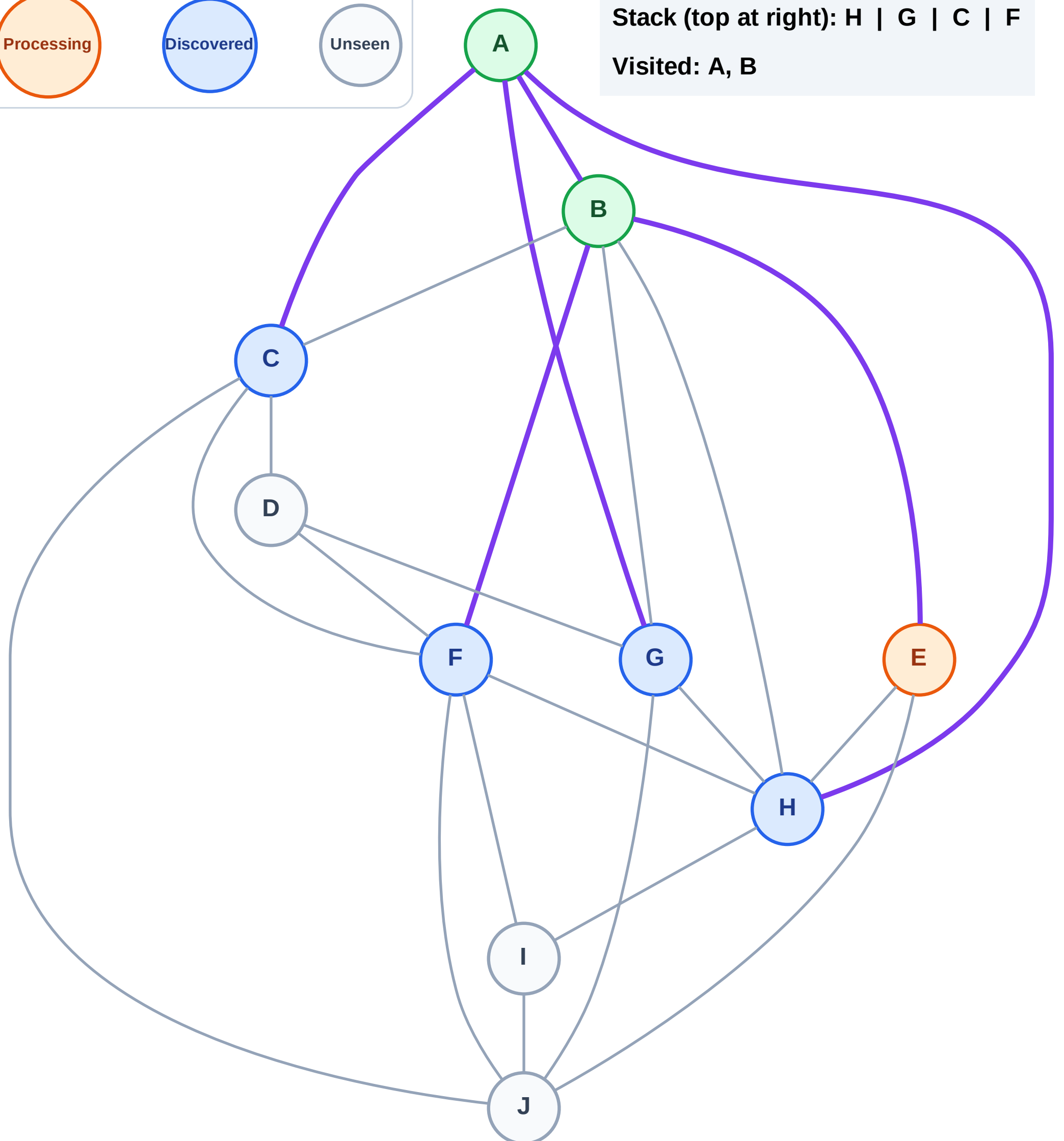
Step 6: Process node E

Legend



Stack (top at right): H | G | C | F

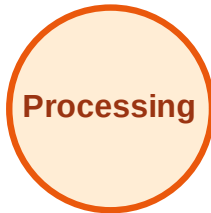
Visited: A, B



DFS Traversal

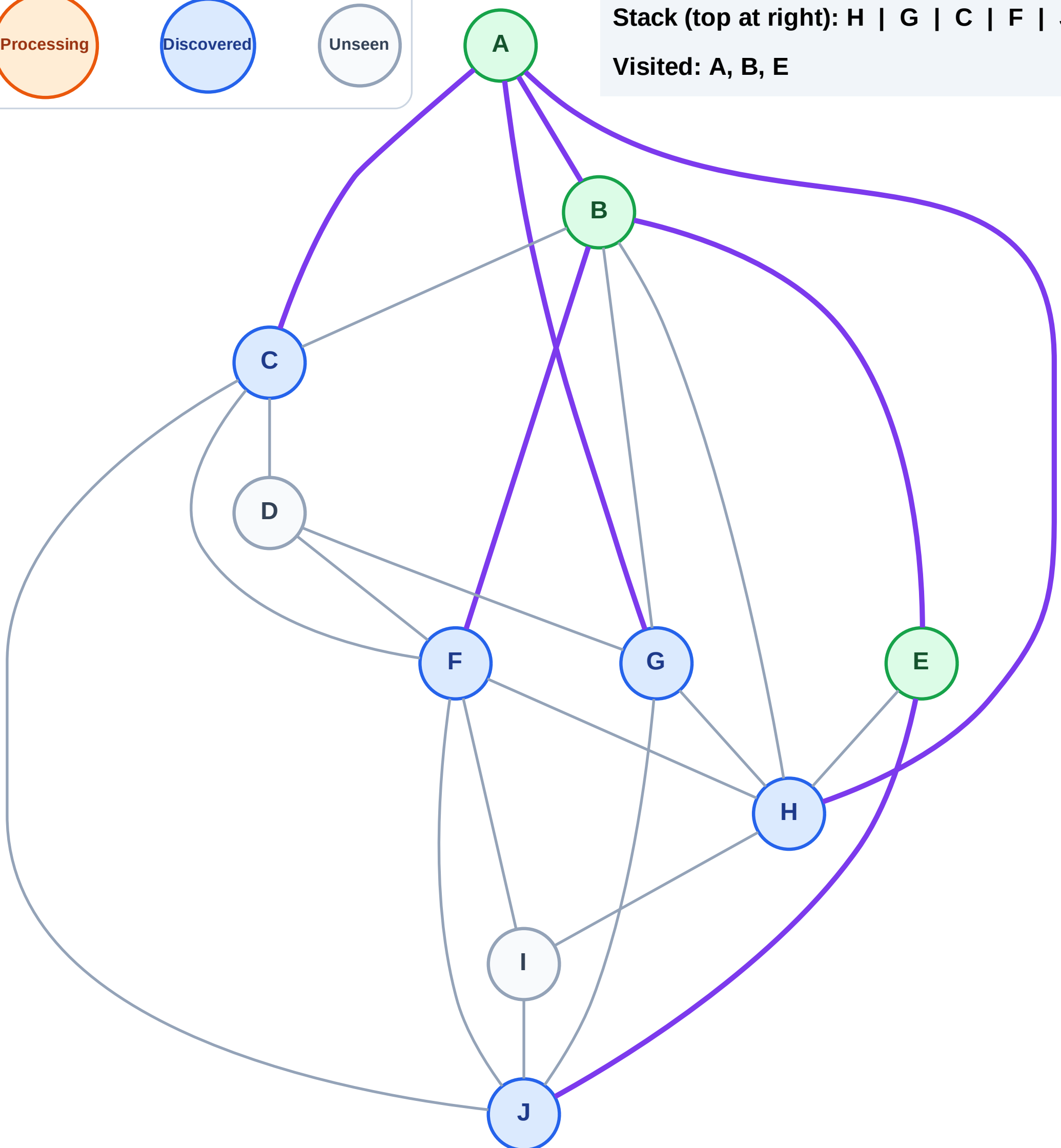
Step 7: Discover J from E

Legend



Stack (top at right): H | G | C | F | J

Visited: A, B, E



DFS Traversal

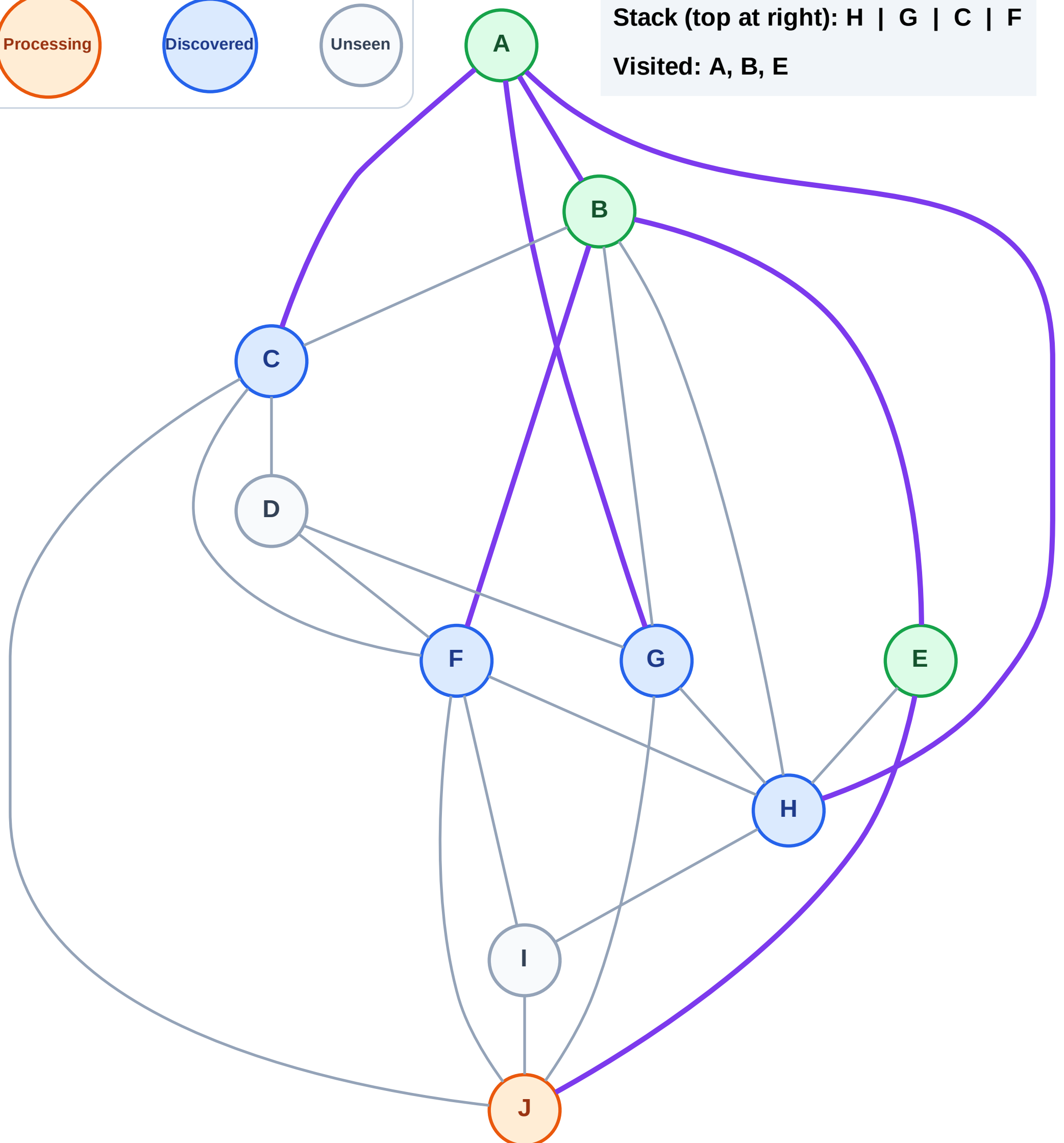
Step 8: Process node J

Legend



Stack (top at right): H | G | C | F

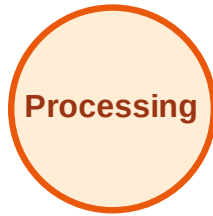
Visited: A, B, E



DFS Traversal

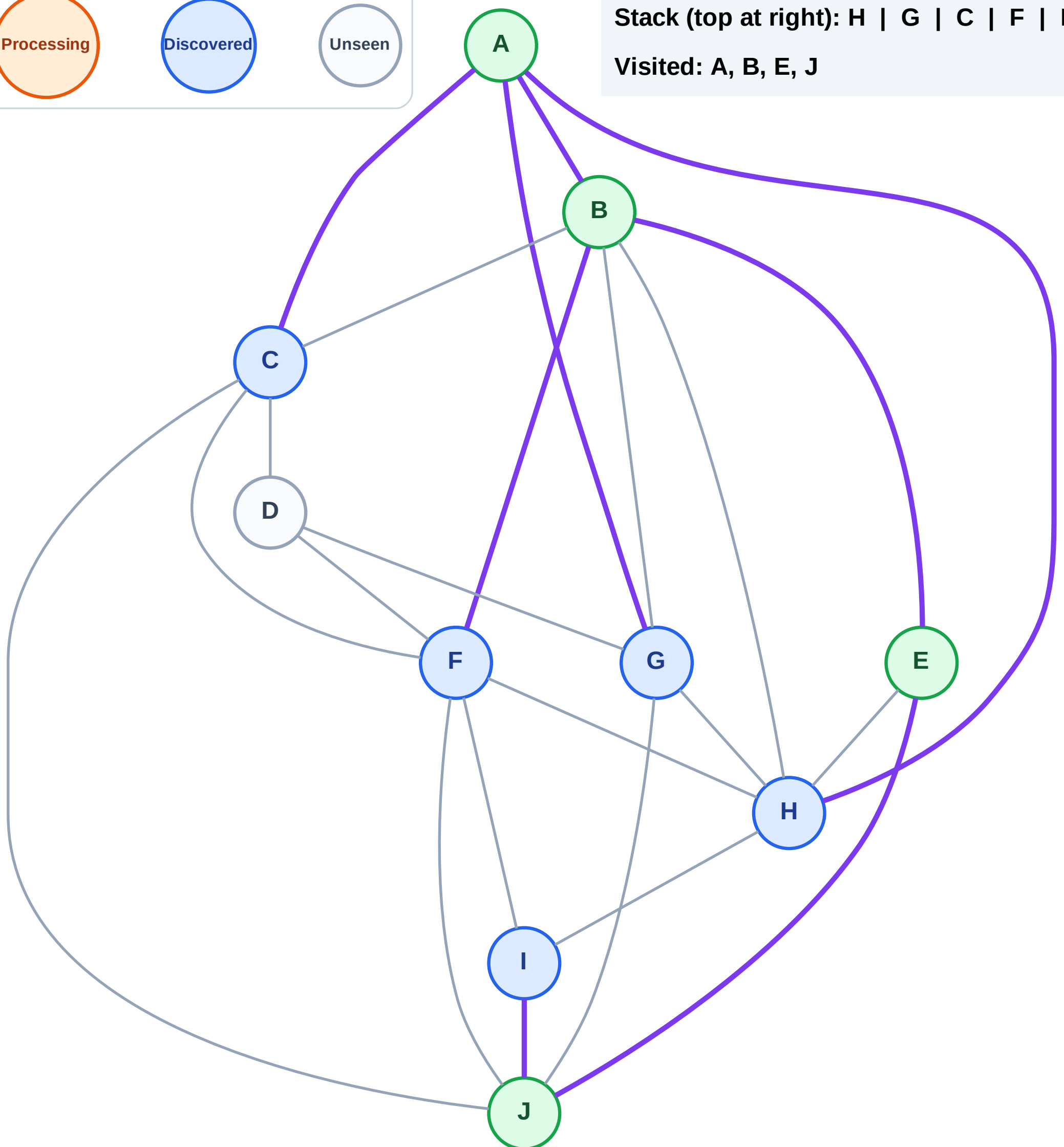
Step 9: Discover I from J

Legend



Stack (top at right): H | G | C | F | I

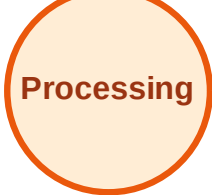
Visited: A, B, E, J



DFS Traversal

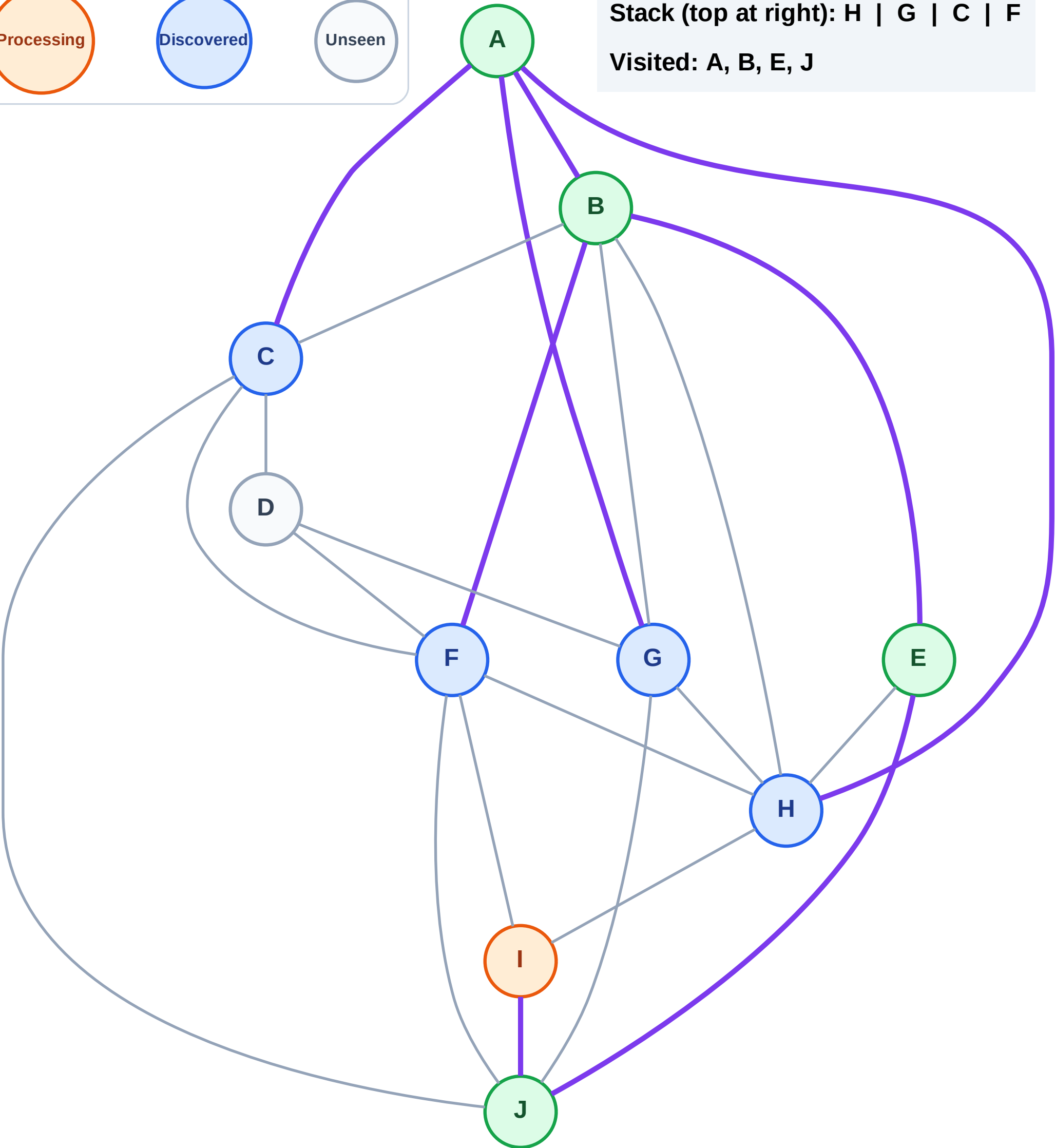
Step 10: Process node I

Legend



Stack (top at right): H | G | C | F

Visited: A, B, E, J



DFS Traversal

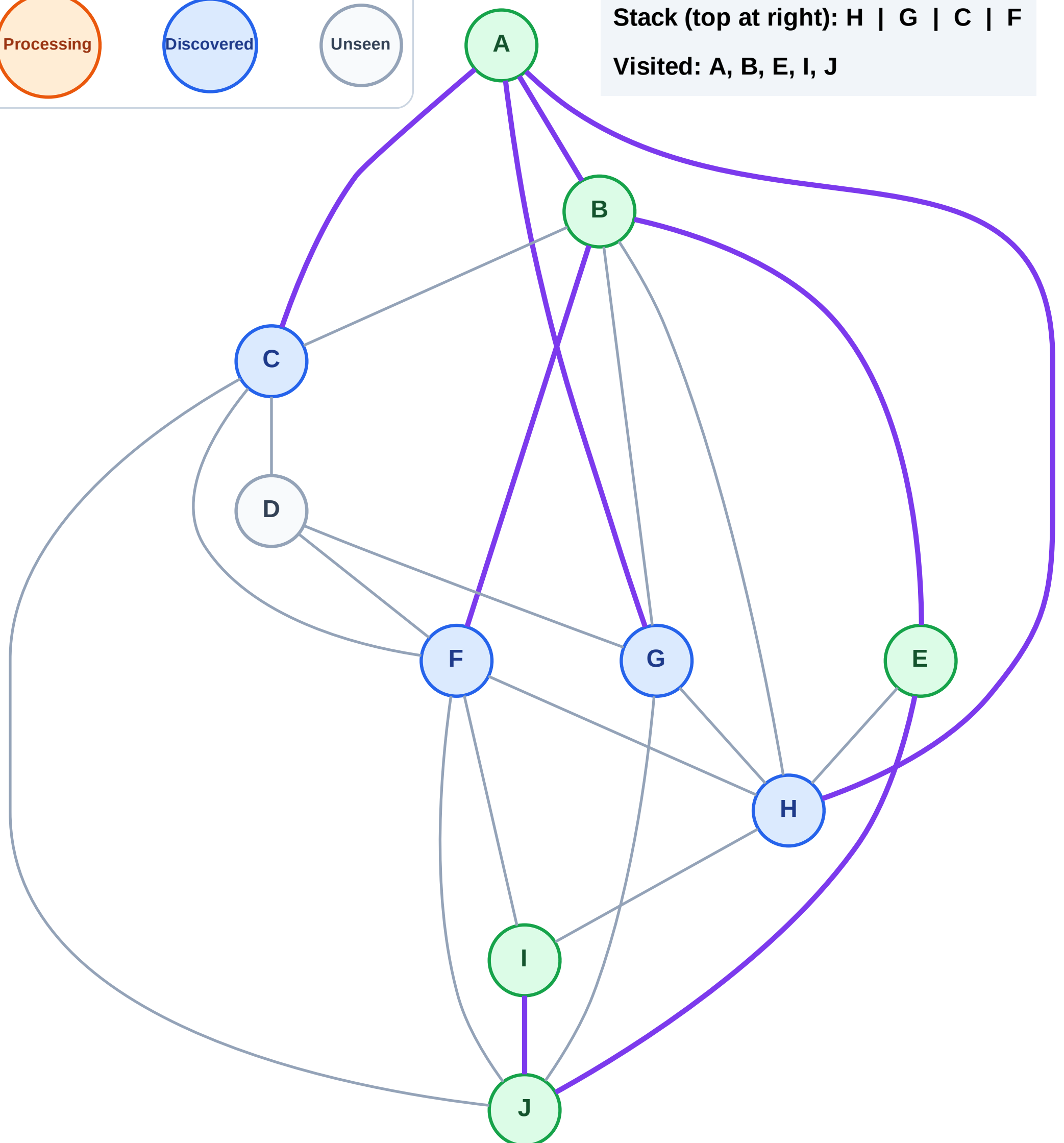
Step 11: No new neighbors from I

Legend



Stack (top at right): H | G | C | F

Visited: A, B, E, I, J



DFS Traversal

Step 12: Process node F

Legend

Complete

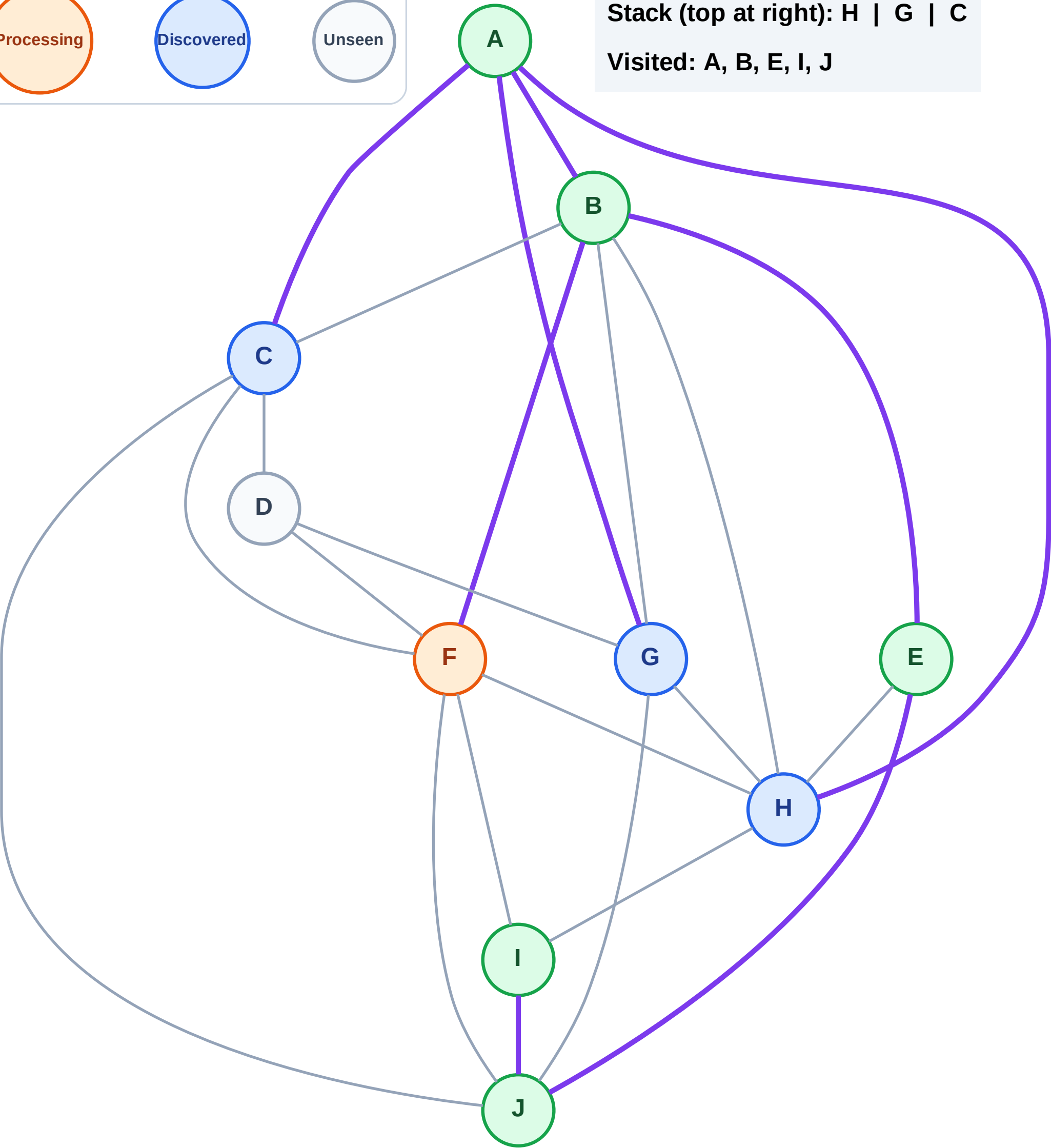
Processing

Discovered

Unseen

Stack (top at right): H | G | C

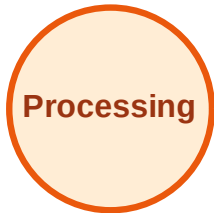
Visited: A, B, E, I, J



DFS Traversal

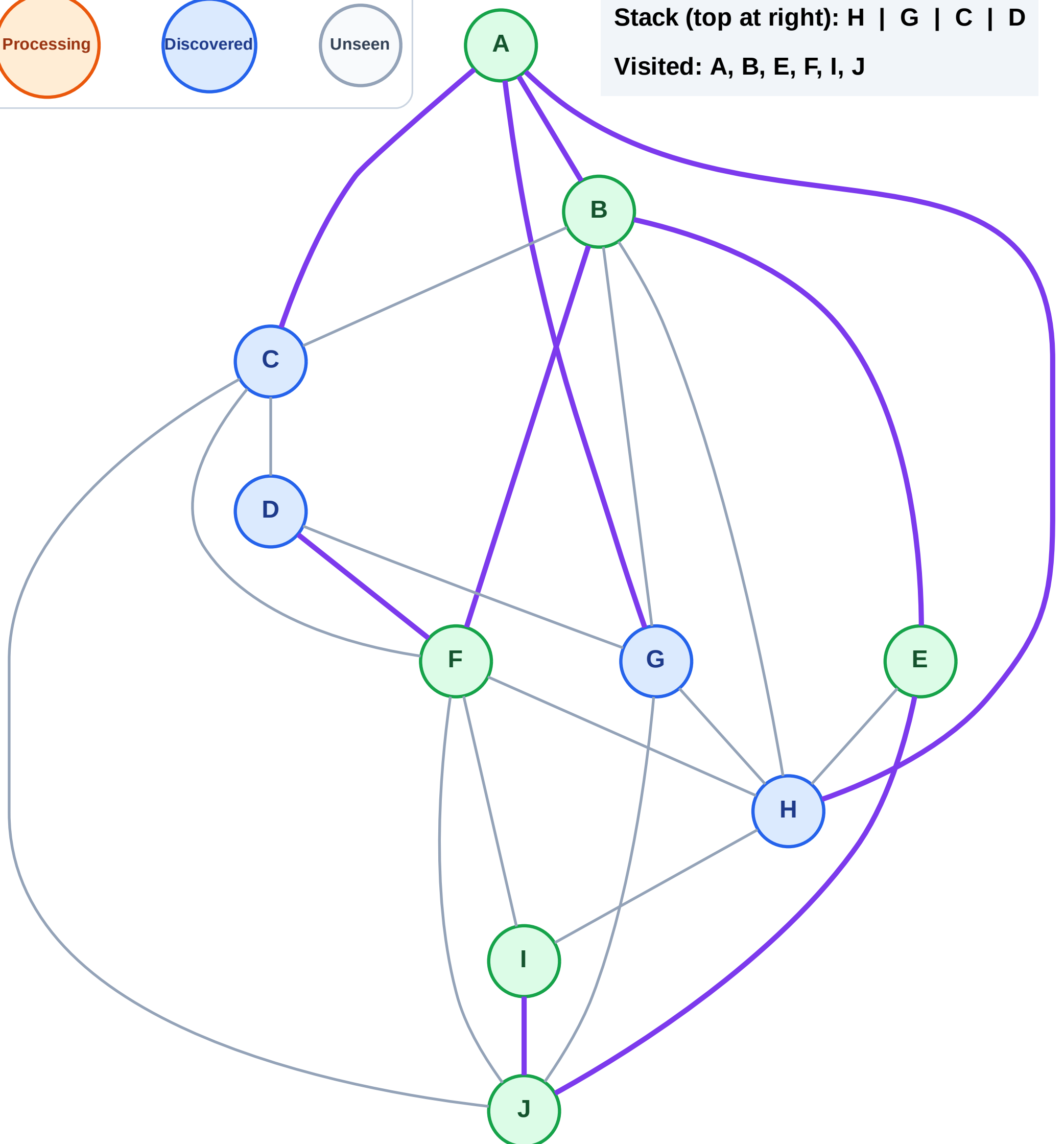
Step 13: Discover D from F

Legend



Stack (top at right): H | G | C | D

Visited: A, B, E, F, I, J



DFS Traversal

Step 14: Process node D

Legend

Complete

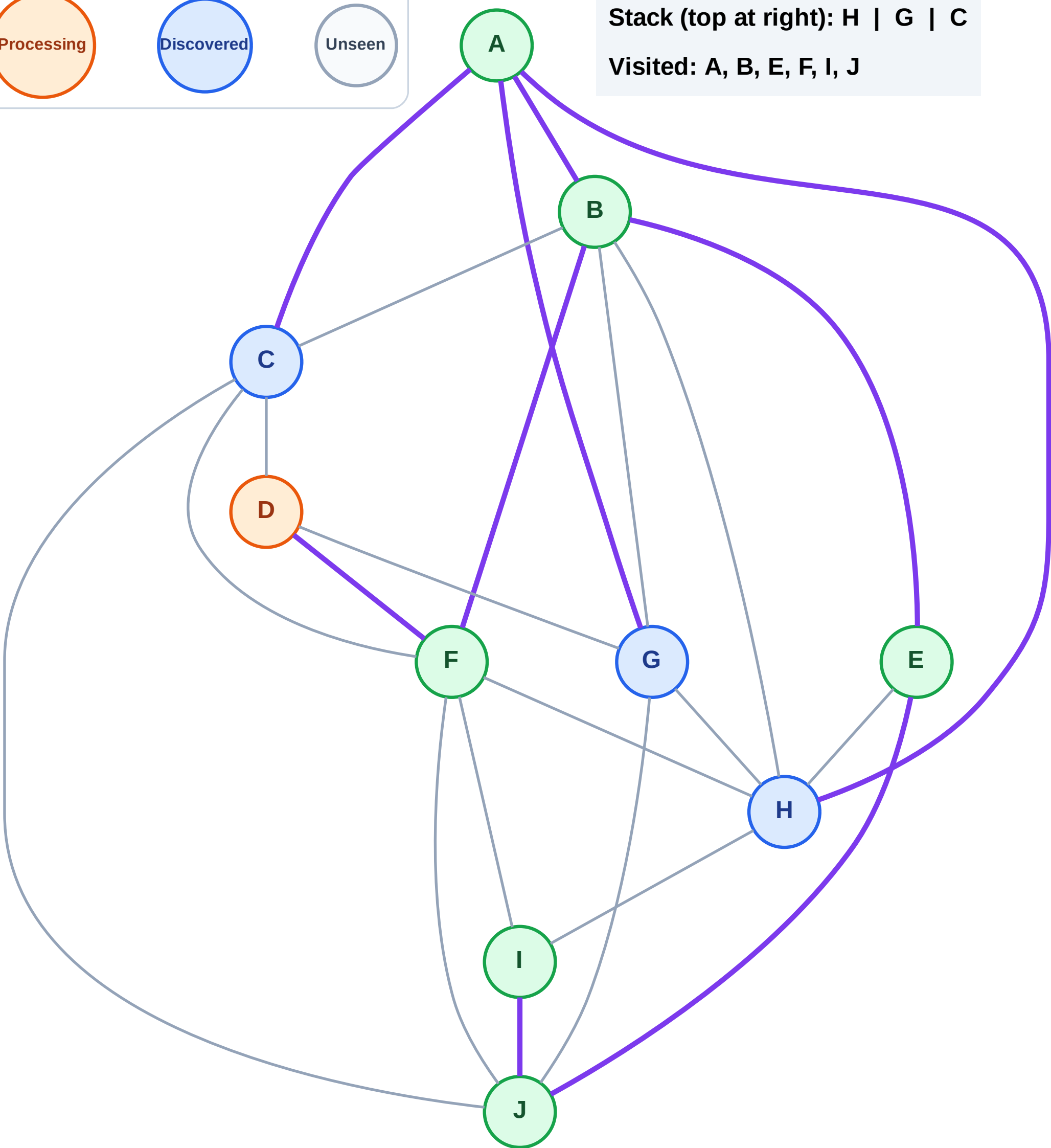
Processing

Discovered

Unseen

Stack (top at right): H | G | C

Visited: A, B, E, F, I, J



DFS Traversal

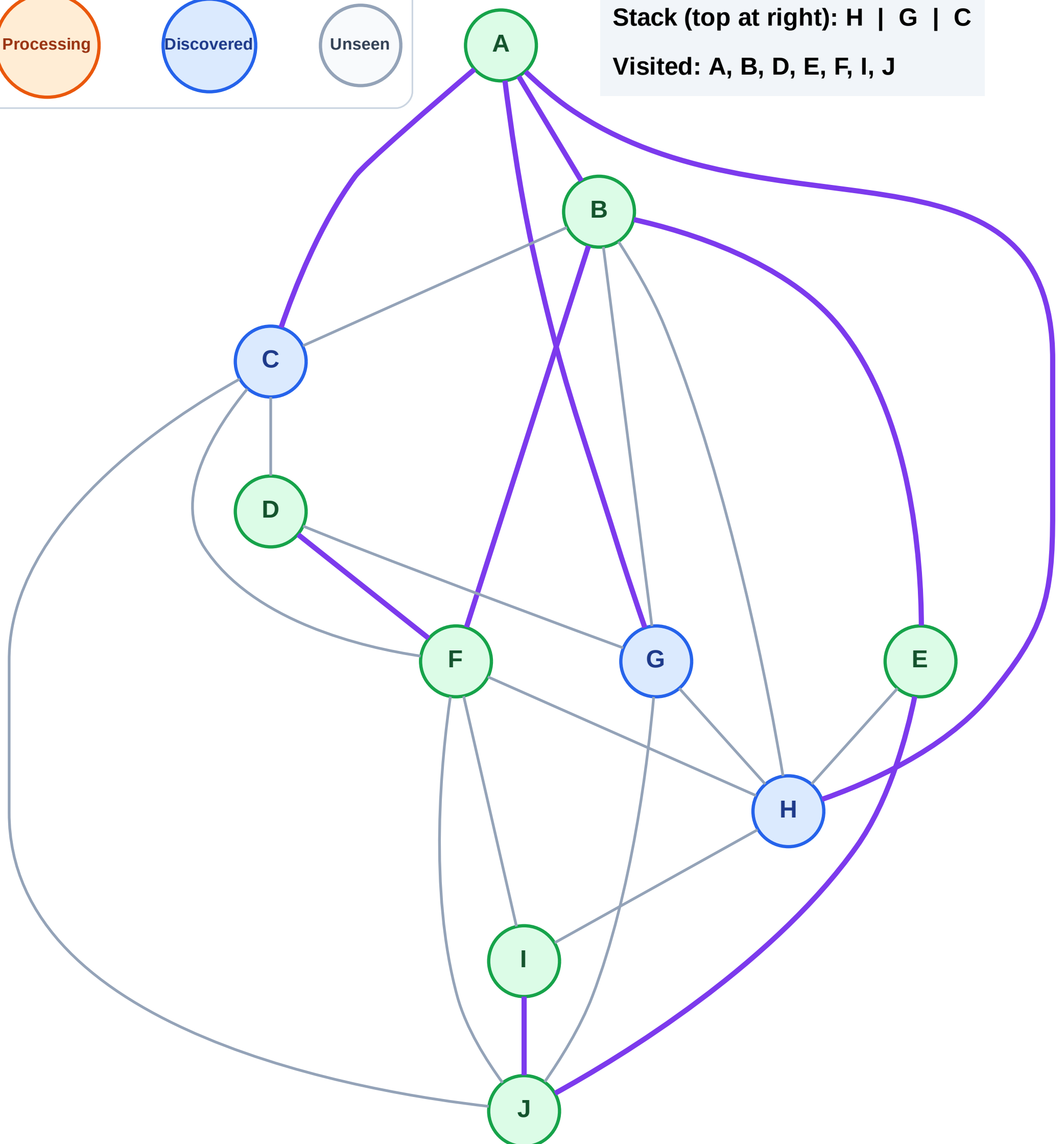
Step 15: No new neighbors from D

Legend



Stack (top at right): H | G | C

Visited: A, B, D, E, F, I, J



DFS Traversal

Step 16: Process node C

Legend

Complete

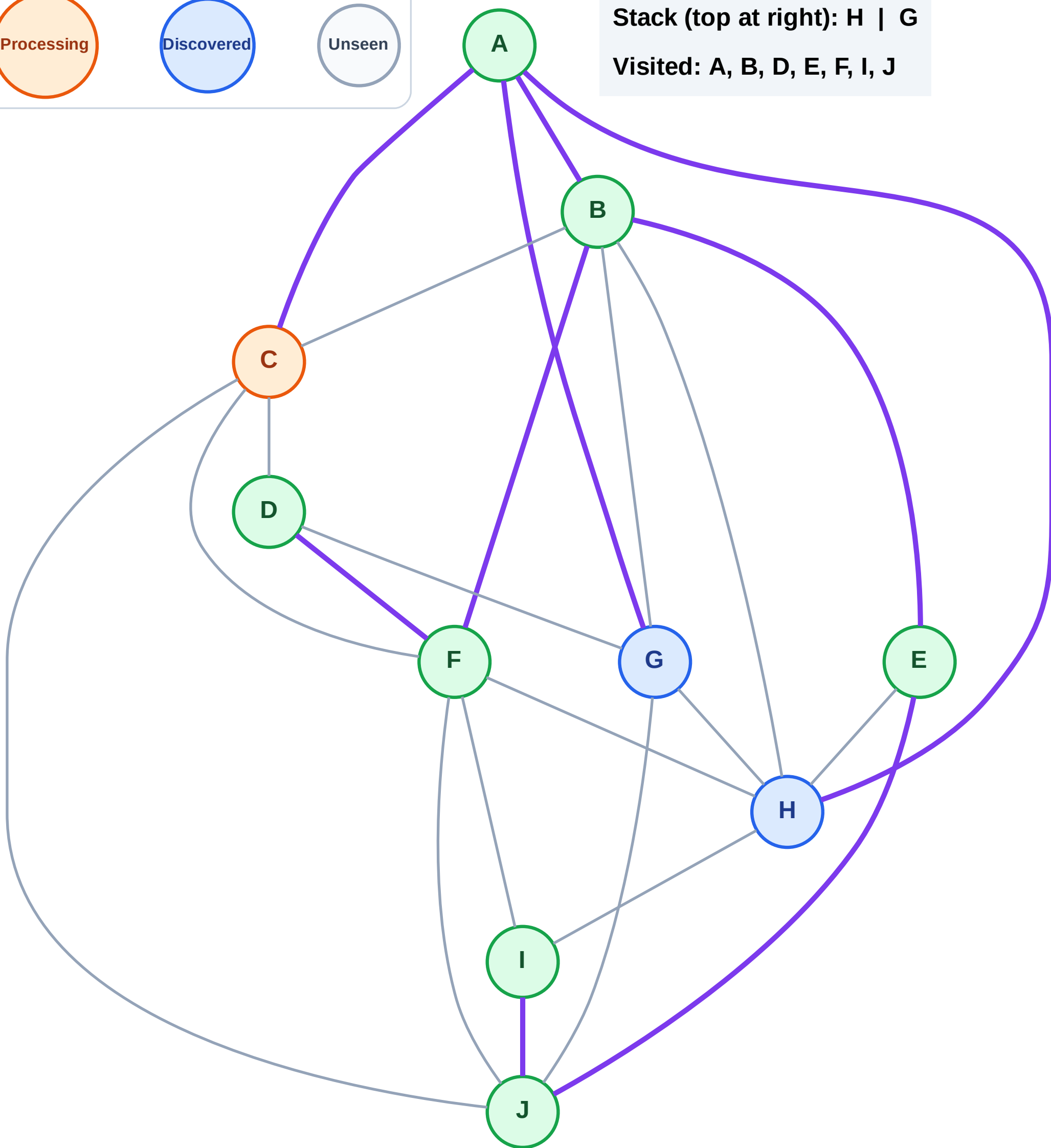
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A, B, D, E, F, I, J



DFS Traversal

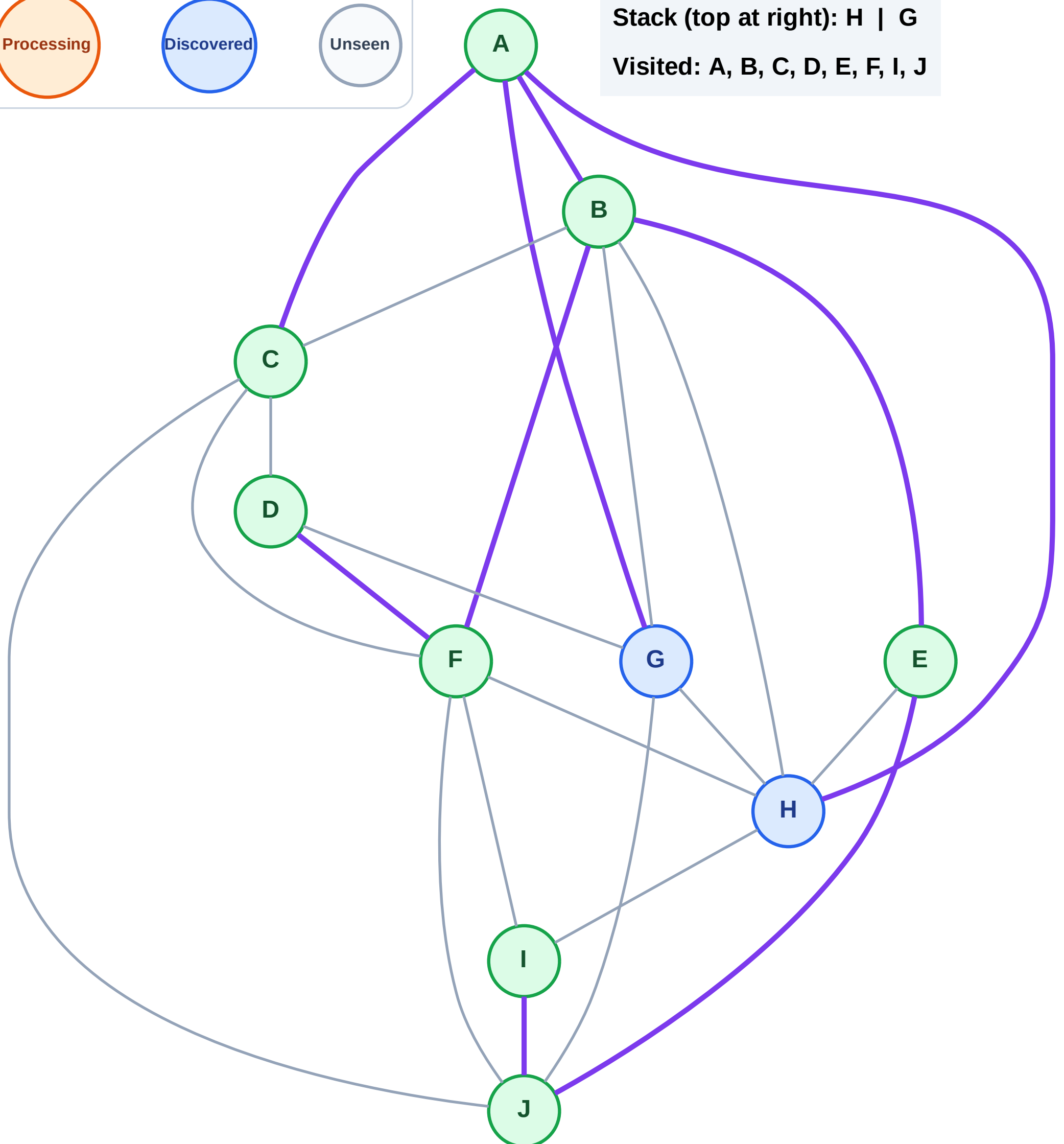
Step 17: No new neighbors from C

Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

Complete

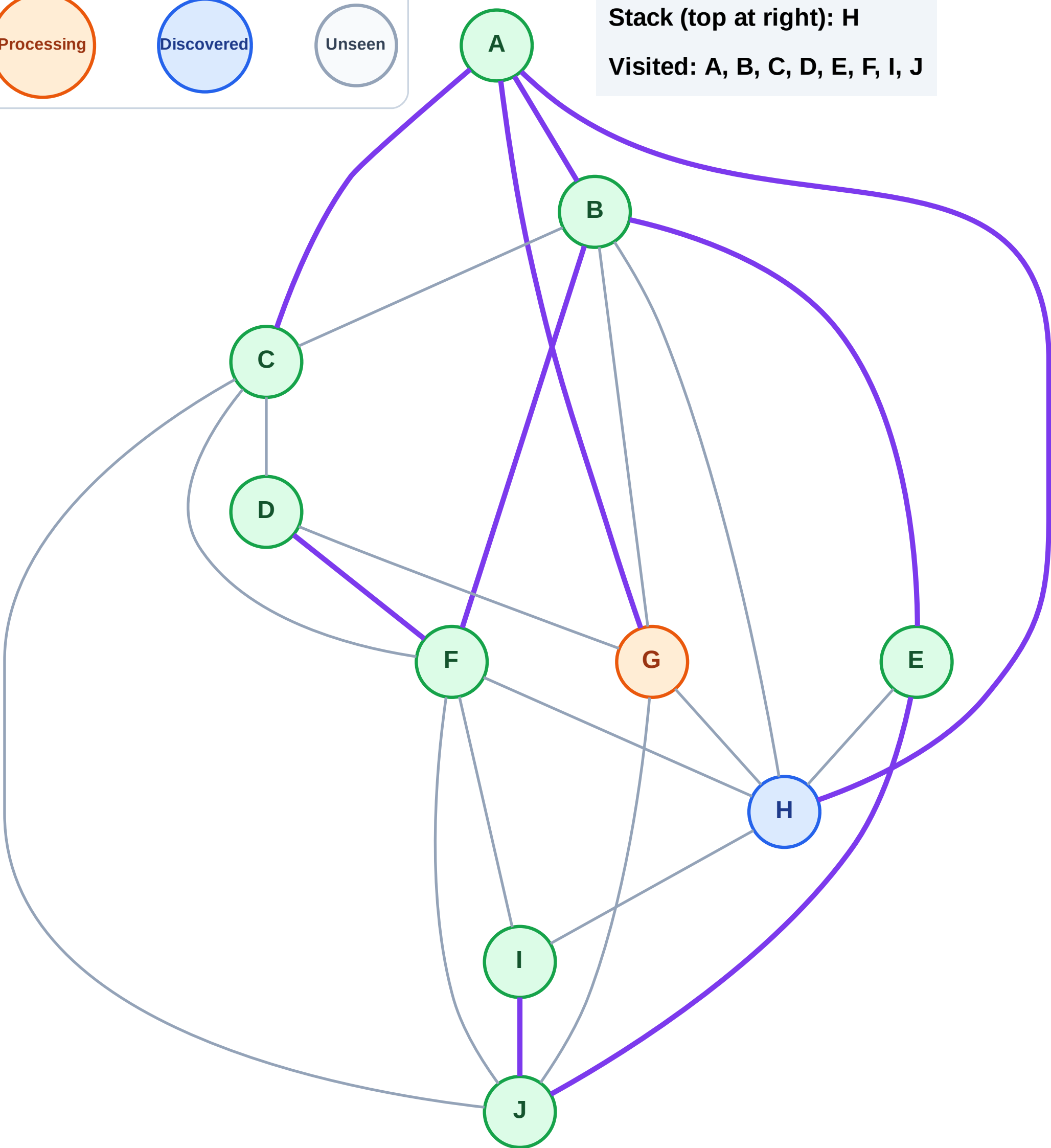
Processing

Discovered

Unseen

Stack (top at right): H

Visited: A, B, C, D, E, F, I, J



DFS Traversal

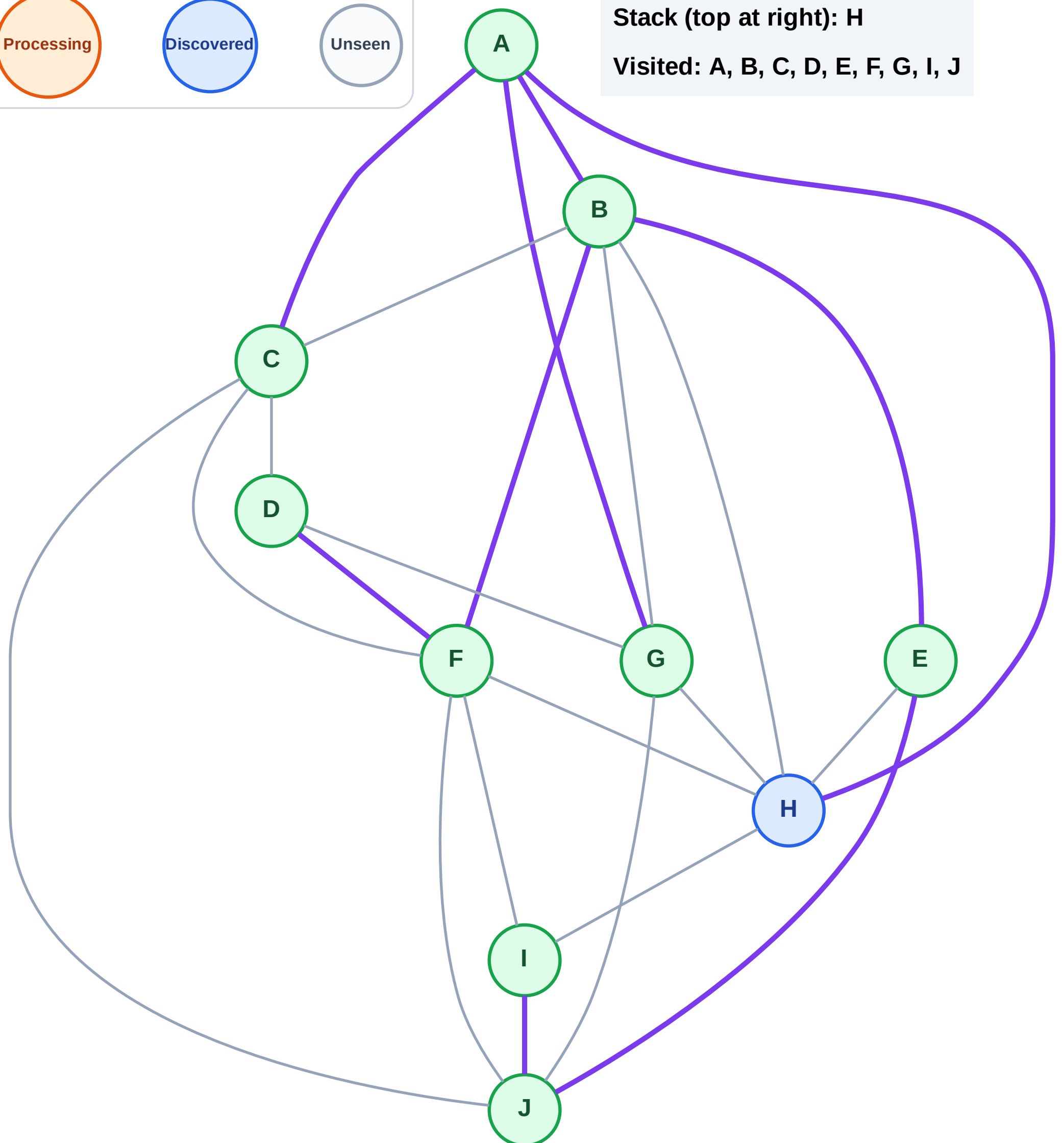
Step 19: No new neighbors from G

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 20: Process node H

Legend

Complete

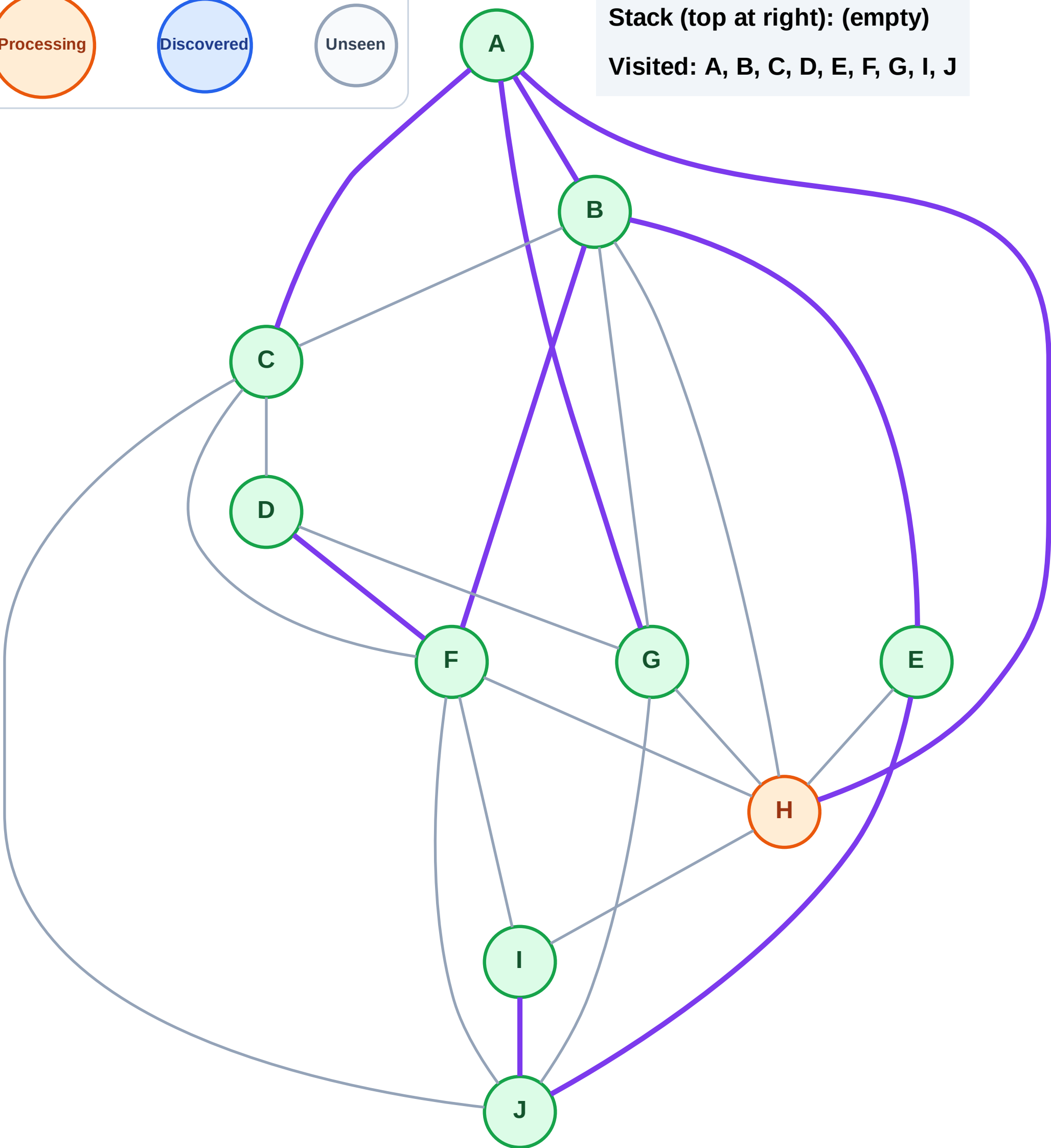
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

